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LYE DRAIN VESSEL

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A167NHM07HP001&002&0003LYE DRAIN VESSELS -7%.pvdb

Date of Analysis : Sep 18, 2026 3:43pm

PV Elite 28, January 2026



LYE DRAIN VESSEL

PV Elite[®] v28



HEXAGON

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3000.000 mm ID
 Vessel Design Length, Tangent to Tangent 12000.00 mm
 Specified Datum Line Distance 0.00 mm
 Internal Design Temperature 120 °C
 Internal Design Pressure 0.350 MPa
 External Design Temperature 120 °C
 External Design Pressure 0.100 MPa
 Maximum Allowable Working Pressure 0.55 MPa
 External Max. Allowable Working Pressure 0.1 MPa
 Shop Test Pressure 0.711 MPa
 Field Test Pressure 0.711 MPa
 Required Minimum Design Metal Temperature 0.0 °C
 Warmest Computed Minimum Design Metal Temperature -48.0 °C
 Wind Design Code No Wind Loads
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 304	...	...	S30400	
Head	SA-240 304	...	...	S30400	
Nozzle	SA-312 TP304	...	...	S30400	
Nozzle	SA-240 304	...	...	S30400	
Nozzle	SA-312 TP304	...	...	S30400	
Re-Pad	SA-240 304	...	...	S30400	
Re-Pad	SA-240 304L	...	...	S30403	
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Rings	SA-240 304	...	...	S30400	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-283 C	...	...	K02401	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
EH-1	0.380	0.10	0.550	0.0000	No	No
SHELL-1	0.380	0.10	0.830	0.0000	N/A	No
SHELL-2	0.380	0.10	0.690	0.0000	N/A	No
SHELL-3	0.380	0.10	0.690	0.0000	N/A	No
SHELL-4	0.380	0.10	0.830	0.0000	N/A	No
Eh-2	0.380	0.10	0.550	0.0000	No	No

Liquid Level: 2400.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.280

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
2400.00	Bar 150.0 x 10.0	Ring R1 Fr20
4850.00	Bar 152.0 x 10.0	Ring R2



7250.00 Bar 152.0 x 10.0
9650.00 Bar 152.0 x 10.0

Ring R1 Fr40
Ring R1 Fr50

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	10.0	8.0	5.3	...	1.00	1.00
Cylinder	3050.0	3000.0	12.0	12.0	5.3	...	1.00	1.00
Cylinder	6050.0	3000.0	10.0	10.0	5.3	...	1.00	1.00
Cylinder	9050.0	3000.0	10.0	10.0	5.3	...	1.00	1.00
Cylinder	11950.0	2900.0	12.0	12.0	5.3	...	1.00	1.00
Ellipse	12000.0	50.0	10.0	8.0	5.3	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	8	7.96323	0.1	0.1
20	Ring	12	9.30505	0.1	0.19
Ring	30	12	8.99572	0.1	0.2
30	Ring	10	8.98888	0.1	0.13
Ring	40	10	8.91062	0.1	0.13
40	Ring	10	8.91062	0.1	0.13
Ring	50	10	8.91062	0.1	0.13
50	Ring	12	8.91739	0.1	0.21
Ring	60	12	9.2282	0.1	0.19
60	70	8	7.96323	0.1	0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	2650	4834.78	No Calc	No Calc
Ring	30	2450	4836.6	5197542	9006961
30	Ring	2450	3115.98	No Calc	No Calc
Ring	40	2400	3115.8	4937822	8497795
40	Ring	2400	3115.8	No Calc	No Calc
Ring	50	2400	3115.8	4868567	8497795
50	Ring	2400	4836.53	No Calc	No Calc
Ring	60	2600	4834.66	5103630	9326621
60	70	No Calc	No Calc	No Calc	No Calc

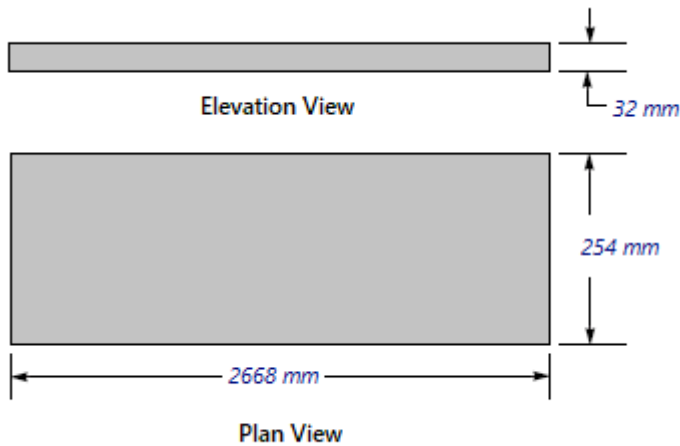
Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-6''	WRC-107/537	0.943	n/a	Passed
N2-3''	WRC-297	0.704	Shell	Passed
N3-6''	WRC-297	0.820	Shell	Passed
N7-4''	WRC-297	0.663	Shell	Passed
N6-4''	WRC-297	0.891	Shell	Passed
N4-3''	WRC-297	0.891	Shell	Passed
N5-6	WRC-297	0.820	Shell	Passed
N8-4''	WRC-297	0.672	Shell	Passed

Foundation/Support Summary**Saddle Parameters:**

Saddle Width	228.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1760.000	mm
Wear Pad Width	280.000	mm
Wear Pad Thickness	10.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	720.000	mm
Baseplate Length	2668.000	mm
Baseplate Thickness	32.000	mm
Baseplate Width	254.000	mm
Number of Ribs (including outside ribs)	4	
Rib Thickness	13.000	mm
Web Thickness	16.000	mm
Height of Center Web	216.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	717727.69	N
Maximum Transverse Saddle Shear Load	56143.07	N
Maximum Longitudinal Saddle Shear Load	112286.13	N

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	534311.12	N
Maximum Transverse Saddle Shear Load	0.00	N
Maximum Longitudinal Saddle Shear Load	0.00	N

Weights:

Fabricated - Bare w/o removable internals	16910.3	kgm
Shop Test - Fabricated + water (full)	108975.3	kgm
Shipping - Fab. + rem. intls.+ shipping app.	16910.3	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	17123.3	kgm
Empty - Fab. + intls. + details + wghts.	17123.3	kgm
Operating - Empty + operating liquid (no ca)	118336.0	kgm
Field Test - Empty weight + test medium (full)	108975.3	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-6''	0.55	OK	...	OK	6.22	OK	Passed
N9-6''	0.83	OK	...	OK	6.22	OK	Passed
N2-3''	0.72	...	...	OK	4.80	OK	No Calc[*]
N3-6''	0.83	OK	...	OK	6.22	OK	Passed
M1-24''	0.66	OK	...	...		OK	Passed
N7-4''	0.83	OK	...	OK	5.26	OK	Passed
N6-4''	0.69	OK	...	OK	5.26	OK	Passed
K1-2''	0.69	...	...	OK	3.42	OK	No Calc[*]
K2-4''	0.57	OK	...	OK	5.26	OK	Passed
K3-2''	0.69	...	...	OK	3.42	OK	No Calc[*]
N4-3''	0.72	...	...	OK	4.80	OK	No Calc[*]
N5-6	0.83	OK	...	OK	6.22	OK	Passed
M2-24''	0.66	OK	...	...		OK	Passed
N8-4''	0.83	OK	...	OK	5.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 0.55 Nozzle : N1-6''
Minimum MAWP Shells/Flanges : 0.55 Element : Eh-2
Minimum MAPnc Shells/Flanges : 0.74 Element : Eh-2

Computed Vessel M.A.W.P. : 0.55 MPa

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N1-6''	0.000	90.000	298.145
20	N9-6''	300.000	90.000	298.145
20	N2-3''	700.000	90.000	138.862
20	N3-6''	1050.000	90.000	298.145
20	M1-24''	1800.000	90.000	1171.200
20	N7-4''	2900.000	90.000	189.662
30	N6-4''	3200.000	90.000	189.662
30	K1-2''	3500.000	90.000	101.270
30	K2-4''	3800.000	90.000	189.662
30	K3-2''	4100.000	90.000	101.270
50	N4-3''	11700.000	90.000	138.862
50	N5-6	11100.000	90.000	298.145
50	M2-24''	10349.999	90.000	1171.200
50	N8-4''	10499.999	270.000	189.662

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
K1-2''	2.000 in	XS	LWN	2.375	5.537	...	...	1500.30	150
K3-2''	2.000 in	XS	LWN	2.375	5.537	...	...	1500.30	150
N2-3''	3.000 in	160	WNF	3.500	11.125	...	...	1502.66	300
N4-3''	3.000 in	160	WNF	3.500	11.125	...	...	1502.66	300
N7-4''	4.000 in	120	WNF	4.500	11.125	200.00	12.00	1503.09	300
N6-4''	4.000 in	120	WNF	4.500	11.125	200.00	10.00	1501.09	300
K2-4''	4.000 in	120	WNF	4.500	11.125	...	...	1501.09	150
N8-4''	4.000 in	120	WNF	4.500	11.125	200.00	10.00	253.09	300
N1-6''	6.000 in	80	WNF	6.625	10.973	300.00	10.00	754.08	300
N9-6''	6.000 in	80	WNF	6.625	10.973	300.00	10.00	1504.36	300
N3-6''	6.000 in	80	WNF	6.625	10.973	300.00	12.00	1504.36	300
N5-6	6.000 in	80	WNF	6.625	10.973	300.00	12.00	1504.36	300
M1-24''	24.000 in Actual		WNF	24.000	12.000	900.00	10.00	1533.29	150
M2-24''	24.000 in Actual		WNF	24.000	12.000	900.00	10.00	1533.29	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1-2''	SA-312 TP304	10.000	10.000	...	...	...
K3-2''	SA-312 TP304	10.000	10.000	...	...	...
N2-3''	SA-312 TP304	10.000	10.000	...	...	...
N4-3''	SA-312 TP304	10.000	10.000	...	...	...
N7-4''	SA-312 TP304	10.000	10.000	8.000	12.000	...
N6-4''	SA-312 TP304	10.000	10.000	10.000	10.000	...
K2-4''	SA-312 TP304	10.000	10.000	...	...	...
N8-4''	SA-312 TP304	10.000	10.000	8.000	10.000	...
N1-6''	SA-312 TP304	8.000	10.000	6.000	10.000	...
N9-6''	SA-312 TP304	10.000	10.000	8.000	10.000	...
N3-6''	SA-312 TP304	10.000	10.000	8.000	12.000	...
N5-6	SA-312 TP304	10.000	10.000	10.000	12.000	...
M1-24''	SA-240 304	10.000	12.000	8.000	10.000	...
M2-24''	SA-240 304	10.000	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1-2''	3500.000	90.0	1490.00	0.00	SHELL-2
K3-2''	4100.000	90.0	1490.00	0.00	SHELL-2
N2-3''	700.000	90.0	1490.00	0.00	SHELL-1
N4-3''	11699.999	90.0	1490.00	0.00	SHELL-4
N7-4''	2900.000	90.0	1490.00	0.00	SHELL-1
N6-4''	3200.000	90.0	1490.00	0.00	SHELL-2
K2-4''	3800.000	90.0	1490.00	0.00	SHELL-2
N8-4''	10499.999	270.0	240.00	0.00	SHELL-4

LYE DRAIN VESSEL

PV Elite 28

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Nozzle Schedule:

Step: 35 3:43pm Sep 18,2026



HEXAGON

N1-6''	...	90.0	720.00	0.00	EH-1
N9-6''	300.000	90.0	1490.00	0.00	SHELL-1
N3-6''	1050.000	90.0	1490.00	0.00	SHELL-1
N5-6	11100.000	90.0	1490.00	0.00	SHELL-4
M1-24''	1800.000	90.0	1490.00	0.00	SHELL-1
M2-24''	10349.999	90.0	1490.00	0.00	SHELL-4

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm THK X 3000.0mm ID X 50.0mm	SA-240 304
1	CYLINDER: 12.0mm THK X 3000.0mm ID X 3000.0mm	SA-240 304
2	CYLINDER: 10.0mm THK X 3000.0mm ID X 3000.0mm	SA-240 304
1	CYLINDER: 12.0mm THK X 3000.0mm ID X 2900.0mm	SA-240 304
2	SADDLE: 228.0mm X 120 DEG	SA-283 C
1	BAR RING STIFFENER: 3024.0mm ID X 3324.0mm OD X 10.0mm	SA-240 304
2	BAR RING STIFFENER: 3020.0mm ID X 3324.0mm OD X 10.0mm	SA-240 304
1	BAR RING STIFFENER: 3024.0mm ID X 3328.0mm OD X 10.0mm	SA-240 304
2	CLASS 150 GR 2.1, 24.0" BLIND FLANGE(S)	SA-182 F304
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
EH-1	[16]	-196						
SHELL-1	[16]	-196						
SHELL-2	[16]	-196						
SHELL-3	[16]	-196						
SHELL-4	[16]	-196						
Eh-2	[16]	-196						
N1-6''	[15]	-196						
N9-6''	[15]	-196						
N2-3''	[15]	-196						
N3-6''	[15]	-196						
M1-24''	[15]	-196						
N7-4''	[15]	-196						
N6-4''	[15]	-196						
K1-2''	[15]	-196						
K2-4''	[15]	-196						
K3-2''	[15]	-196						
N4-3''	[15]	-196						
N5-6	[15]	-196						
M2-24''	[15]	-196						
N8-4''	[15]	-196						
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -48

Required Minimum Design Metal Temperature 0.0 °C

Warmest Computed Minimum Design Metal Temperature -48.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

LYE DRAIN VESSEL

Design Internal Pressure (for Hydrotest)	0.35	MPa
Design Internal Temperature	120.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	0.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code No Wind Loads

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg
M.A.W.P + Static Head to Nozzle		Y
Consider MAP New and Cold in Noz. Design		N
Consider External Loads for Nozzle Des.		Y
Use ASME VIII-1 Appendix 1-9		N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	EH-1
Distance "FROM" to "TO"	50 mm
Inside Diameter	3000 mm
Element Thickness	8 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm



Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	120	°C
Design External Pressure	0.1	MPa
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	108.2	MPa
Allowable Stress, Hydrotest	184.5	MPa
Material Density	0.008027	kg/cm³
P Number Thickness	0	mm
Yield Stress, Operating	162.8	MPa
External Pressure Chart Name	HA-1	
UNS Number	S30400	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2400	mm
Liquid Density	0.001278	kg/cm³

Element From Node	10	
Detail Type	Nozzle	
Detail ID	N1-6''	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	672.32	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	10	
Detail Type	Weight	
Detail ID	dip pie and elb	
Dist. from "FROM" Node / Offset dist	-550	mm
Miscellaneous Weight	686.42	N
Offset from Element Centerline	750	mm

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	SHELL-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3000	mm
Element Thickness	12	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	120	°C
Design External Pressure	0.1	MPa
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



Element From Node	20	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	680	mm
Width of Saddle	228	mm
Height of Saddle at Bottom	1760	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	280	mm
Thickness of Wear Plate	10	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2400	mm
Liquid Density	0.001278	kg/cm ³
Element From Node	20	
Detail Type	Ring	
Detail ID	Ring R1 Fr20	
Dist. from "FROM" Node / Offset dist	2350	mm
Inside Diameter of Ring	3024	mm
Thickness of Ring	10	mm
Outside Diameter of Ring	3324	mm
Material Name	SA-240 304	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N9-6''	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	964.96	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2-3''	
Dist. from "FROM" Node / Offset dist	650	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	388.83	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N3-6''	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	



Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	970.79 N
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element From Node	20
Detail Type	Nozzle
Detail ID	M1-24''
Dist. from "FROM" Node / Offset dist	1750 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	6078.3 N
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-240 304

Element From Node	20
Detail Type	Nozzle
Detail ID	N7-4''
Dist. from "FROM" Node / Offset dist	2850 mm
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	570.23 N
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element From Node	20
Detail Type	Weight
Detail ID	DAVIT-1
Dist. from "FROM" Node / Offset dist	1750 mm
Miscellaneous Weight	490.3 N
Offset from Element Centerline	1700 mm

Element From Node	20
Detail Type	Weight
Detail ID	LL-1
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	2157.3 N
Offset from Element Centerline	1400 mm

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	SHELL-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3000 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	120 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2400	mm
Liquid Density	0.001278	kg/cm ³

Element From Node	30	
Detail Type	Ring	
Detail ID	Ring R2	
Dist. from "FROM" Node / Offset dist	1800	mm
Inside Diameter of Ring	3020	mm
Thickness of Ring	10	mm
Outside Diameter of Ring	3324	mm
Material Name	SA-240 304	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N6-4''	
Dist. from "FROM" Node / Offset dist	150	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	569.72	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K1-2''	
Dist. from "FROM" Node / Offset dist	450	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XS	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	147.33	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K2-4''	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	481.53	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K3-2''	
Dist. from "FROM" Node / Offset dist	1050	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XS	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	147.33	N
Grade of Attached Flange	GR 2.1	



Nozzle Matl	SA-312 TP304
Element From Node	30
Detail Type	Weight
Detail ID	STILLING WELL
Dist. from "FROM" Node / Offset dist	750 mm
Miscellaneous Weight	568.76 N
Offset from Element Centerline	1500 mm

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	SHELL-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3000 mm
Element Thickness	10 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	120 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2400 mm
Liquid Density	0.001278 kg/cm³

Element From Node	40
Detail Type	Ring
Detail ID	Ring R1 Fr40
Dist. from "FROM" Node / Offset dist	1200 mm
Inside Diameter of Ring	3020 mm
Thickness of Ring	10 mm
Outside Diameter of Ring	3324 mm
Material Name	SA-240 304
Height of Section Ring	0 mm
Using Custom Stiffener Section	No

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	SHELL-4
Distance "FROM" to "TO"	2900 mm
Inside Diameter	3000 mm
Element Thickness	12 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.35 MPa
Design Temperature Internal Pressure	120 °C
Design External Pressure	0.1 MPa
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304



Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 2190 mm
 Width of Saddle 228 mm
 Height of Saddle at Bottom 1760 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 280 mm
 Thickness of Wear Plate 10 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.6000002
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid 50
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 2400 mm
 Liquid Density 0.001278 kg/cm³

Element From Node 50
 Detail Type Ring
 Detail ID Ring R1 Fr50
 Dist. from "FROM" Node / Offset dist 600 mm
 Inside Diameter of Ring 3024 mm
 Thickness of Ring 10 mm
 Outside Diameter of Ring 3328 mm
 Material Name SA-240 304
 Height of Section Ring 0 mm
 Using Custom Stiffener Section No

Element From Node 50
 Detail Type Nozzle
 Detail ID N4-3''
 Dist. from "FROM" Node / Offset dist 2650 mm
 Nozzle Diameter 3 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 388.83 N
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-312 TP304

Element From Node 50
 Detail Type Nozzle
 Detail ID N5-6
 Dist. from "FROM" Node / Offset dist 2050 mm
 Nozzle Diameter 6 in.
 Nozzle Schedule 80
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 810.59 N
 Grade of Attached Flange GR 2.1
 Nozzle Matl SA-312 TP304

Element From Node 50
 Detail Type Nozzle
 Detail ID M2-24''



Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	6132.5	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-240	304

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N8-4''	
Dist. from "FROM" Node / Offset dist	1450	mm
Nozzle Diameter	4	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	255.58	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312	TP304

Element From Node	50	
Detail Type	Weight	
Detail ID	DAVIT-2	
Dist. from "FROM" Node / Offset dist	1300	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	1650	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	DIP pipe	
Dist. from "FROM" Node / Offset dist	2050	mm
Miscellaneous Weight	833.52	N
Offset from Element Centerline	1200	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	LL-2	
Dist. from "FROM" Node / Offset dist	1400	mm
Miscellaneous Weight	2157.3	N
Offset from Element Centerline	1400	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Eh-2	
Distance "FROM" to "TO"	50	mm
Inside Diameter	3000	mm
Element Thickness	8	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.35	MPa
Design Temperature Internal Pressure	120	°C
Design External Pressure	0.1	MPa
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240	304
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
-------------------	----	--



Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2400 mm
Liquid Density	0.001278 kg/cm ³

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
EH-1		50	...	50	...
SHELL-1		3050	...	3000	...
SHELL-2		6050	...	3000	...
SHELL-3		9050	...	3000	...
SHELL-4		11950	...	2900	...
Eh-2		12000	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	EH-1	0.3801	10	...	3000	108.2	1
	SHELL-1	0.3801	12	...	3000	108.2	2
	SHELL-2	0.3801	10	...	3000	108.2	3
	SHELL-3	0.3801	10	...	3000	108.2	4
	SHELL-4	0.3801	12	...	3000	108.2	5
	Eh-2	0.3801	10	...	3000	108.2	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
	EH-1	0.35	0.55	0.74	8	5.27141	1
	SHELL-1	0.35	0.83	1.1	12	5.28069	2
	SHELL-2	0.35	0.69	0.92	10	5.28069	3
	SHELL-3	0.35	0.69	0.92	10	5.28069	4
	SHELL-4	0.35	0.83	1.1	12	5.28069	5
	Eh-2	0.35	0.55	0.74	8	5.27141	6

Minimum 0.55 0.74

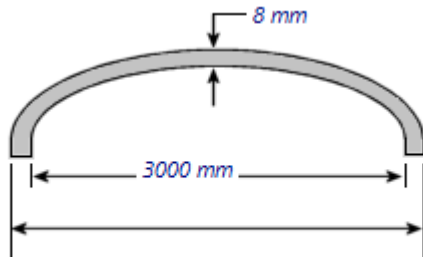
MAWP: 0.55 MPa, limited by: Eh-2.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: EH-1

Elliptical Head Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		50.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		71.3 MPa
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		10360759.0 mm²	SG of Contents		1.280
Element Volume		3888.4 ltr	Weight of Contents		43703.6 N
Empty Weight		8907.8 N	Operating Weight		52611.4 N
			Comp. All. Ope. Stress		59.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.38 \cdot 3000 \cdot 1) / (2 \cdot 108 \cdot 1 - 0.2 \cdot 0.38)$$

$$= 5.2714 + 0.0000 = 5.2714 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 108 \cdot 1 \cdot 8) / (1 \cdot 3000 + 0.2 \cdot 8)$$

$$= 0.58 - 0.03 = 0.55 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8) / (1 \cdot 3000 + 0.2 \cdot 8)$$

$$= 0.74 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.38 \cdot (1 \cdot 3000 + 0.2 \cdot 8)) / (2 \cdot 1 \cdot 8)$$

$$= 71.309 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) + 0$
 $= 5.281 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.030 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (108 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10)$
 $= 0.72 - 0.03 = 0.69 \text{ MPa}$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.456 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

EXTERNAL PRESSURE RESULTS for: EH-1

Elliptical Head From 10 to 20 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.000	3016.00	377.00	0.0003684	34.25

$$MAEP = B / (K0 \cdot Do / t) = 34.2 / (0.9 \cdot 377) = 0.1 \text{ MPa}$$

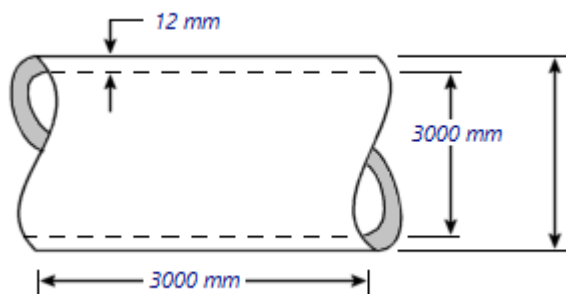
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.963	3016.00	378.74	0.0003667	34.09

$$MAEP = B / (K0 \cdot Do / t) = 34.1 / (0.9 \cdot 379) = 0.1 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		3000.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		47.7 MPa
Spec. Min. Thk t _e , or s _a		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		28500530.0 mm²	SG of Contents		1.280
Element Volume		21209.6 ltr	Weight of Contents		228172.4 N
Empty Weight		26814.5 N	Operating Weight		254987.0 N
			Comp. All. Ope. Stress		68.3 MPa
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) \\
 &= 5.2807 + 0.0000 = 5.2807 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (108 \cdot 1 \cdot 12) / (1500 + 0.6 \cdot 12) \\
 &= 0.86 - 0.03 = 0.83 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 12) / (1500 + 0.6 \cdot 12) \\
 &= 1.1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.38 \cdot (1500 + 0.6 \cdot 12)) / (1 \cdot 12) \\
 &= 47.742 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f) * (1 - R_f / R_o) 0.398 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: SHELL-1

Cylindrical Shell From 20 to Ring R1 Fr20 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	2650.00	252.00	0.8763	0.0003799	35.31

$$MAEP = (4*B)/(3*(Do/t)) = (4*35.3)/(3*252) = 0.19 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.305	3024.00	2650.00	324.98	0.8763	0.0002622	24.38

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.4)/(3*325) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	4834.78	252.00	1.5988	0.0002035	18.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9)/(3*252) = 0.1 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr20 to 30 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	2450.00	252.00	0.8102	0.0004120	38.30

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.3)/(3*252) = 0.2 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.996	3024.00	2450.00	336.16	0.8102	0.0002712	25.21

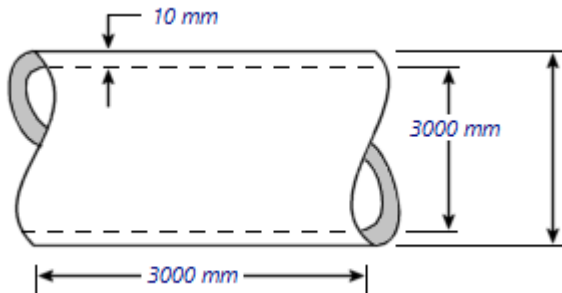
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.2)/(3*336) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	4836.60	252.00	1.5994	0.0002034	18.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9)/(3*252) = 0.1 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		3000.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		57.2 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		28462830.0 mm ²	SG of Contents		1.280
Element Volume		21209.6 ltr	Weight of Contents		228172.4 N
Empty Weight		22330.4 N	Operating Weight		250502.9 N
			Comp. All. Ope. Stress		64.4 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) \\
 &= 5.2807 + 0.0000 = 5.2807 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (108 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10) \\
 &= 0.72 - 0.03 = 0.69 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10) \\
 &= 0.92 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.38 \cdot (1500 + 0.6 \cdot 10)) / (1 \cdot 10) \\
 &= 57.245 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f) * (1 - R_f / R_o) 0.332 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cylindrical Shell From 30 to Ring R2 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2450.00	302.00	0.8113	0.0003136	29.15

$$MAEP = (4*B)/(3*(Do/t)) = (4*29.2)/(3*302) = 0.13 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.989	3020.00	2450.00	335.97	0.8113	0.0002711	25.20

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.2)/(3*336) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	3115.98	302.00	1.0318	0.0002438	22.67

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*302) = 0.1 \text{ MPa}$$

Cylindrical Shell From Ring R2 to 40 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2400.00	302.00	0.7947	0.0003204	29.79

$$MAEP = (4*B)/(3*(Do/t)) = (4*29.8)/(3*302) = 0.13 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.911	3020.00	2400.00	338.92	0.7947	0.0002735	25.42

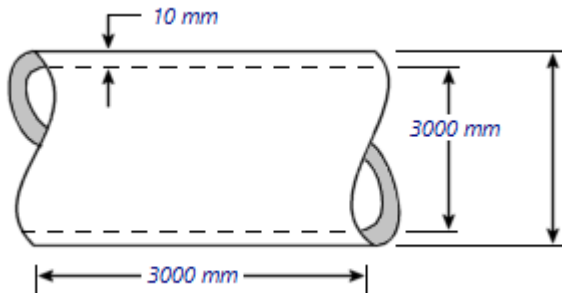
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.4)/(3*339) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	3115.80	302.00	1.0317	0.0002438	22.67

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*302) = 0.1 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		3000.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		57.2 MPa
Spec. Min. Thk t _e , or s _a		10.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		28462830.0 mm ²	SG of Contents		1.280
Element Volume		21209.6 ltr	Weight of Contents		228172.4 N
Empty Weight		22330.4 N	Operating Weight		250502.9 N
			Comp. All. Ope. Stress		64.4 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) \\
 &= 5.2807 + 0.0000 = 5.2807 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (108 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10) \\
 &= 0.72 - 0.03 = 0.69 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10) \\
 &= 0.92 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.38 \cdot (1500 + 0.6 \cdot 10)) / (1 \cdot 10) \\
 &= 57.245 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f) * (1 - R_f / R_o) 0.332 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: SHELL-3

Cylindrical Shell From 40 to Ring R1 Fr40 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2400.00	302.00	0.7947	0.0003204	29.79

$$MAEP = (4*B)/(3*(Do/t)) = (4*29.8)/(3*302) = 0.13 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.911	3020.00	2400.00	338.92	0.7947	0.0002735	25.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.4)/(3*339) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	3115.80	302.00	1.0317	0.0002438	22.67

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*302) = 0.1 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr40 to 50 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	2400.00	302.00	0.7947	0.0003204	29.79

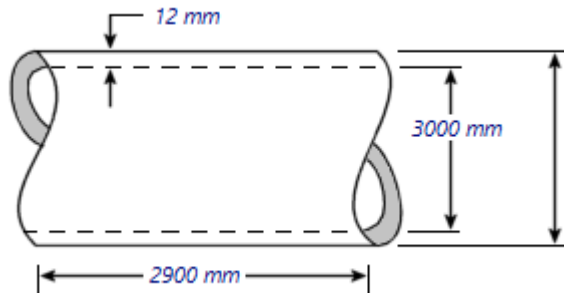
$$MAEP = (4*B)/(3*(Do/t)) = (4*29.8)/(3*302) = 0.13 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.911	3020.00	2400.00	338.92	0.7947	0.0002735	25.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.4)/(3*339) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.000	3020.00	3115.80	302.00	1.0317	0.0002438	22.67

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.7)/(3*302) = 0.1 \text{ MPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: SHELL-4**

Cylindrical Shell Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		2900.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		47.7 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		27550514.0 mm²	SG of Contents		1.280
Element Volume		20502.6 ltr	Weight of Contents		220566.7 N
Empty Weight		25920.8 N	Operating Weight		246487.5 N
			Comp. All. Ope. Stress		68.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) \\
 &= 5.2807 + 0.0000 = 5.2807 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (108 \cdot 1 \cdot 12) / (1500 + 0.6 \cdot 12) \\
 &= 0.86 - 0.03 = 0.83 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 12) / (1500 + 0.6 \cdot 12) \\
 &= 1.1 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.38 \cdot (1500 + 0.6 \cdot 12)) / (1 \cdot 12) \\
 &= 47.742 \text{ MPa}
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f) * (1 - R_f / R_o) 0.398 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: SHELL-4

Cylindrical Shell From 50 to Ring R1 Fr50 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	2400.00	252.00	0.7937	0.0004210	39.14

$$MAEP = (4*B)/(3*(Do/t)) = (4*39.1)/(3*252) = 0.21 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.917	3024.00	2400.00	339.11	0.7937	0.0002736	25.43

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.4)/(3*339) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	4836.53	252.00	1.5994	0.0002034	18.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9)/(3*252) = 0.1 \text{ MPa}$$

Cylindrical Shell From Ring R1 Fr50 to 60 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	2600.00	252.00	0.8598	0.0003875	36.02

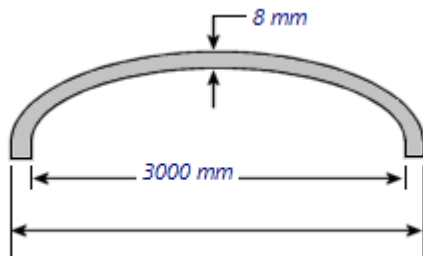
$$MAEP = (4*B)/(3*(Do/t)) = (4*36)/(3*252) = 0.19 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.228	3024.00	2600.00	327.69	0.8598	0.0002644	24.58

$$MAEP = (4*B)/(3*(Do/t)) = (4*24.6)/(3*328) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	3024.00	4834.66	252.00	1.5988	0.0002035	18.92

$$MAEP = (4*B)/(3*(Do/t)) = (4*18.9)/(3*252) = 0.1 \text{ MPa}$$

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Eh-2**

Elliptical Head Design Information:					
Design Pressure	P	0.380 MPa	Int. Design Temperature		120 °C
Material		SA-240 304	Ext. Design Temperature		120 °C
Length		50.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	108.2 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3000.000 mm	Actual Stress		71.3 MPa
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		10360759.0 mm²	SG of Contents		1.280
Element Volume		3888.4 ltr	Weight of Contents		43703.6 N
Empty Weight		8907.8 N	Operating Weight		52611.4 N
			Comp. All. Ope. Stress		68.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.38 \cdot 3000 \cdot 1) / (2 \cdot 108 \cdot 1 - 0.2 \cdot 0.38)$$

$$= 5.2714 + 0.0000 = 5.2714 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.030 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 108 \cdot 1 \cdot 8) / (1 \cdot 3000 + 0.2 \cdot 8)$$

$$= 0.58 - 0.03 = 0.55 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 8) / (1 \cdot 3000 + 0.2 \cdot 8)$$

$$= 0.74 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.38 \cdot (1 \cdot 3000 + 0.2 \cdot 8)) / (2 \cdot 1 \cdot 8)$$

$$= 71.309 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (0.38 \cdot 1500) / (108 \cdot 1 - 0.6 \cdot 0.38) + 0$
 $= 5.281 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.030 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (108 \cdot 1 \cdot 10) / (1500 + 0.6 \cdot 10)$
 $= 0.72 - 0.03 = 0.69 \text{ MPa}$

% Elongation per Table UG-79-1 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.456 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm

EXTERNAL PRESSURE RESULTS for: Eh-2

Elliptical Head From 60 to 70 Ext. Chart: HA-1 at 120 °C

Elastic Modulus from Chart: HA-1 at 120 °C: 185917 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
8.000	3016.00	377.00	0.0003684	34.25

$$MAEP = B / (K_0 \cdot Do / t) = 34.2 / (0.9 \cdot 377) = 0.1 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
7.963	3016.00	378.74	0.0003667	34.09

$$MAEP = B / (K_0 \cdot Do / t) = 34.1 / (0.9 \cdot 379) = 0.1 \text{ MPa}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.711 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.455 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.927 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.601 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.500 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.500 MPa
App 27-4	= M.A.W.P.	0.547 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 0.55 • 1

= 0.711 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	138.8	184.5	0.752	0.74
SHELL-1	93.0	184.5	0.504	0.74
SHELL-2	111.5	184.5	0.604	0.74
SHELL-3	111.5	184.5	0.604	0.74
SHELL-4	93.0	184.5	0.504	0.74
Eh-2	138.8	184.5	0.752	0.74

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-6''	Nozzle	117.00	114.60	1.021
N1-6''	Pad	138.00	108.20	1.275
N9-6''	Nozzle	117.00	114.60	1.021
N9-6''	Pad	138.00	108.20	1.275
N2-3''	Nozzle	117.00	114.60	1.021
N3-6''	Nozzle	117.00	114.60	1.021
N3-6''	Pad	115.00	115.00	1.000
M1-24''	Nozzle	138.00	108.20	1.275
M1-24''	Pad	138.00	108.20	1.275
N7-4''	Nozzle	138.00	134.60	1.025
N7-4''	Pad	115.00	115.00	1.000
N6-4''	Nozzle	117.00	114.60	1.021
N6-4''	Pad	115.00	115.00	1.000
K1-2''	Nozzle	117.00	114.60	1.021
K2-4''	Nozzle	117.00	114.60	1.021
K3-2''	Nozzle	117.00	114.60	1.021
N4-3''	Nozzle	117.00	114.60	1.021
N5-6	Nozzle	117.00	114.60	1.021
N5-6	Pad	115.00	115.00	1.000
M2-24''	Nozzle	138.00	108.20	1.275
M2-24''	Pad	138.00	108.20	1.275
N8-4''	Nozzle	138.00	134.60	1.025
N8-4''	Pad	115.00	115.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	108.20	1.275



Ring R2	138.00	108.20	1.275
Ring R1 Fr40	138.00	108.20	1.275
Ring R1 Fr50	138.00	108.20	1.275

Minimum 1.275

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	108.20	1.275
SHELL-1	138.00	108.20	1.275
SHELL-2	138.00	108.20	1.275
SHELL-3	138.00	108.20	1.275
SHELL-4	138.00	108.20	1.275
Eh-2	138.00	108.20	1.275

Minimum 1.275

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1-6''	6.19	152.10	0.041
N9-6''	6.19	152.10	0.041
N2-3''	3.08	152.10	0.020
N3-6''	6.19	152.10	0.041
M1-24''	18.50	179.40	0.103
N7-4''	4.05	179.40	0.023
N6-4''	4.05	152.10	0.027
K1-2''	4.31	152.10	0.028
K2-4''	4.05	152.10	0.027
K3-2''	4.31	152.10	0.028
N4-3''	3.08	152.10	0.020
N5-6	6.19	152.10	0.041
M2-24''	18.50	179.40	0.103
N8-4''	4.05	179.40	0.023

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b) = $1.30 * \text{M.A.W.P.} * \text{Sa/S}$ 0.711 MPa
 UG99b(b)(35) = $1.30 * \text{Design Pres} * \text{Sa/S}$ 0.455 MPa
 UG99(c) = $1.30 * \text{M.A.P.} - \text{Head(Hyd)}$ 0.927 MPa
 UG100 = $1.10 * \text{M.A.W.P.} * \text{Sa/S}$ 0.601 MPa
 PED = $\max(1.43*DP, 1.25*DP*\text{ratio})$ 0.500 MPa
 PED&UG99(b)(35) = $\max(1.43*DP, 1.3*DP*\text{ratio})$ 0.500 MPa
 App 27-4 = M.A.W.P. 0.547 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = $1.3 * 0.55 * 1$
 = 0.711 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
EH-1	138.8	184.5	0.752	0.74
SHELL-1	93.0	184.5	0.504	0.74
SHELL-2	111.5	184.5	0.604	0.74
SHELL-3	111.5	184.5	0.604	0.74
SHELL-4	93.0	184.5	0.504	0.74
Eh-2	138.8	184.5	0.752	0.74

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-6''	Nozzle	117.00	114.60	1.021
N1-6''	Pad	138.00	108.20	1.275
N9-6''	Nozzle	117.00	114.60	1.021
N9-6''	Pad	138.00	108.20	1.275
N2-3''	Nozzle	117.00	114.60	1.021
N3-6''	Nozzle	117.00	114.60	1.021
N3-6''	Pad	115.00	115.00	1.000
M1-24''	Nozzle	138.00	108.20	1.275
M1-24''	Pad	138.00	108.20	1.275
N7-4''	Nozzle	138.00	134.60	1.025
N7-4''	Pad	115.00	115.00	1.000
N6-4''	Nozzle	117.00	114.60	1.021
N6-4''	Pad	115.00	115.00	1.000
K1-2''	Nozzle	117.00	114.60	1.021
K2-4''	Nozzle	117.00	114.60	1.021
K3-2''	Nozzle	117.00	114.60	1.021
N4-3''	Nozzle	117.00	114.60	1.021
N5-6	Nozzle	117.00	114.60	1.021
N5-6	Pad	115.00	115.00	1.000
M2-24''	Nozzle	138.00	108.20	1.275
M2-24''	Pad	138.00	108.20	1.275
N8-4''	Nozzle	138.00	134.60	1.025
N8-4''	Pad	115.00	115.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
Ring R1 Fr20	138.00	108.20	1.275

Ring R2	138.00	108.20	1.275
Ring R1 Fr40	138.00	108.20	1.275
Ring R1 Fr50	138.00	108.20	1.275

Minimum 1.275

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
EH-1	138.00	108.20	1.275
SHELL-1	138.00	108.20	1.275
SHELL-2	138.00	108.20	1.275
SHELL-3	138.00	108.20	1.275
SHELL-4	138.00	108.20	1.275
Eh-2	138.00	108.20	1.275

Minimum 1.275

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1-6''	6.19	152.10	0.041
N9-6''	6.19	152.10	0.041
N2-3''	3.08	152.10	0.020
N3-6''	6.19	152.10	0.041
M1-24''	18.50	179.40	0.103
N7-4''	4.05	179.40	0.023
N6-4''	4.05	152.10	0.027
K1-2''	4.31	152.10	0.028
K2-4''	4.05	152.10	0.027
K3-2''	4.31	152.10	0.028
N4-3''	3.08	152.10	0.020
N5-6	6.19	152.10	0.041
M2-24''	18.50	179.40	0.103
N8-4''	4.05	179.40	0.023

Elements Suitable for Test Pressure.

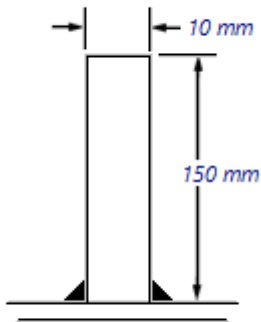
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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

Stiffening Ring Calculations for: Ring R1 Fr20, SA-240 304 , Bar Ring: 150.0 x 10.0 mm

Material treatment options:



Effective Length of Shell: 210 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2514.522	6.0000	15087.133
Ring :	1500.000	87.0000	130500.062
Total:	4014.523		145587.188

Centroid of Ring plus Shell: 36 mm

	Inertia	Distance	A*Dist ²
Shell:	30174.234	30.2651	2303248.000
Ring :	2812499.750	-50.7349	3861044.750
Total:	2842674.000		6164293.000

Available Moment of Inertia, Ring + Shell: 0.901E+07 mm⁴

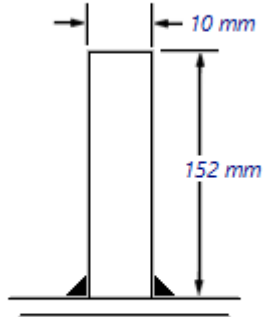
Required Stress in Ring plus Shell Breq 18.02 MPa
 Required Strain in Ring plus Shell Areq 0.0001930

Required Moment of Inertia, Ring plus Shell:

$$\begin{aligned}
 &= (D_o^2 \cdot L_s(t + A_s/L_s)A)/10.9 \\
 &= (3024^2 \cdot 2550(12 + 1500/2550)0.000193)/10.9 \\
 &= 5197542 \text{ mm}^4
 \end{aligned}$$

Stiffening Ring Calculations for: Ring R2, SA-240 304 , Bar Ring: 152.0 x 10.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	1519.999	86.0000	130719.867
Total:	3431.595		140277.844

Centroid of Ring plus Shell: 41 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	35.8783	2460709.000
Ring :	2926499.500	-45.1216	3094661.750
Total:	2942429.250		5555371.000

Available Moment of Inertia, Ring + Shell: 0.850E+07 mm⁴

Required Stress in Ring plus Shell Breq 21.31 MPa
Required Strain in Ring plus Shell Areq 0.0002290

Required Moment of Inertia, Ring plus Shell:

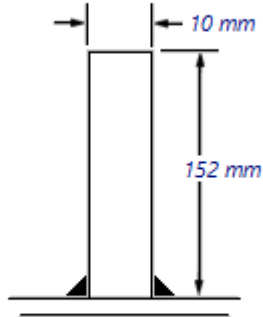
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020^2 \cdot 2425(10 + 1520/2425)0.000228999) / 10.9$$

$$= 4937823 \text{ mm}^4$$

Stiffening Ring Calculations for: Ring R1 Fr40, SA-240 304 , Bar Ring: 152.0 x 10.0 mm

Material treatment options:



Effective Length of Shell: 191 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	1911.596	5.0000	9557.980
Ring :	1519.999	86.0000	130719.867
Total:	3431.595		140277.844

Centroid of Ring plus Shell: 41 mm

	Inertia	Distance	A*Dist ²
Shell:	15929.951	35.8783	2460709.000
Ring :	2926499.500	-45.1216	3094661.750
Total:	2942429.250		5555371.000

Available Moment of Inertia, Ring + Shell: 0.850E+07 mm⁴

Required Stress in Ring plus Shell Breq 21.30 MPa
Required Strain in Ring plus Shell Areq 0.0002280

Required Moment of Inertia, Ring plus Shell:

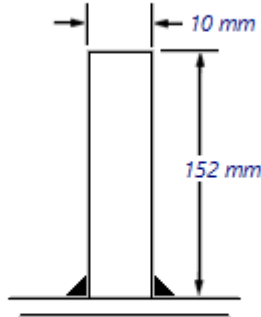
$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3020^2 \cdot 2400(10 + 1520/2400)0.000227999) / 10.9$$

$$= 4868567 \text{ mm}^4$$

Stiffening Ring Calculations for: Ring R1 Fr50, SA-240 304 , Bar Ring: 152.0 x 10.0 mm

Material treatment options:



Effective Length of Shell: 210 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	2514.522	6.0000	15087.133
Ring :	1520.001	88.0001	133760.188
Total:	4034.523		148847.312

Centroid of Ring plus Shell: 37 mm

	Inertia	Distance	A*Dist ²
Shell:	30174.234	30.8934	2399867.250
Ring :	2926510.750	-51.1066	3970075.500
Total:	2956685.000		6369943.000

Available Moment of Inertia, Ring + Shell: 0.933E+07 mm⁴

Required Stress in Ring plus Shell Breq 17.99 MPa
 Required Strain in Ring plus Shell Areq 0.0001930

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (3024^2 \cdot 2500(12 + 1520/2500)0.000193) / 10.9$$

$$= 5103631 \text{ mm}^4$$

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	3016	8	0.00036841	34.2466
20	Ring	2650	3024	12	0.00037989	35.3144
Ring	30	2450	3024	12	0.00041204	38.3028
30	Ring	2450	3020	10	0.00031361	29.1531
Ring	40	2400	3020	10	0.00032045	29.7886
40	Ring	2400	3020	10	0.00032045	29.7886
Ring	50	2400	3020	10	0.00032045	29.7886
50	Ring	2400	3024	12	0.000421	39.1358
Ring	60	2600	3024	12	0.00038746	36.0177
60	70	No Calc	3016	8	0.00036841	34.2466

External Pressure Calculations:

From	To	External Actual T.	External Required T.	External Design Pressure	External M.A.W.P.
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		mm	mm	MPa	MPa
10	20	8	7.96323	0.1	0.1
20	Ring	12	9.30505	0.1	0.19
Ring	30	12	8.99572	0.1	0.2
30	Ring	10	8.98888	0.1	0.13
Ring	40	10	8.91062	0.1	0.13
40	Ring	10	8.91062	0.1	0.13
Ring	50	10	8.91062	0.1	0.13
50	Ring	12	8.91739	0.1	0.21
Ring	60	12	9.2282	0.1	0.19
60	70	8	7.96323	0.1	0.1

Minimum

0.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	Ring	2650	4834.78	No Calc	No Calc
Ring	30	2450	4836.6	5197542	9006961
30	Ring	2450	3115.98	No Calc	No Calc
Ring	40	2400	3115.8	4937822	8497795
40	Ring	2400	3115.8	No Calc	No Calc
Ring	50	2400	3115.8	4868567	8497795
50	Ring	2400	4836.53	No Calc	No Calc
Ring	60	2600	4834.66	5103630	9326621
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	908.403	3888.42	908.403	3888.42	63.5882
20	30	2734.5	21209.6	2734.5	21209.6	191.415
30	40	2277.22	21209.6	2277.22	21209.6	159.406
40	50	2277.22	21209.6	2277.22	21209.6	159.406
50	60	2643.36	20502.6	2643.36	20502.6	185.035
60	70	908.403	3888.42	908.403	3888.42	63.5882
Total		11749	91908.06	11749	91908.06	822

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	4456.82	-250	300	...	Liquid: 10
10	Nozl	73.3616	-649.519	750	...	N1-6''
10	Wght	70	-649.519	750	...	dip pie and elb
20	Sadl	626.268	680	1619	...	Left Saddle
20	Liqd	23268.7	1500	300	...	Liquid 20
20	Ring	128.47	2350	...	...	Ring R1 Fr20
20	Nozl	105.293	250	1584.14	...	N9-6''
20	Nozl	42.4279	650	1544.45	...	N2-3''
20	Nozl	105.93	1000	1584.14	...	N3-6''
20	Nozl	663.247	1750	1804.8	...	M1-24''
20	Nozl	62.2216	2850	1557.15	...	N7-4''
20	Wght	50	1750	1700	...	DAVIT-1
20	Wght	220	1500	1212.44	700	LL-1
30	Liqd	23268.7	1500	300	...	Liquid 30
30	Ring	130.101	1800	...	...	Ring R2
30	Nozl	62.1666	150	1557.15	...	N6-4''
30	Nozl	16.0763	450	1530.16	...	K1-2''
30	Nozl	52.5425	750	1557.15	...	K2-4''
30	Nozl	16.0763	1050	1530.16	...	K3-2''
30	Wght	58.001	750	1500	...	STILLING WELL
40	Liqd	23268.7	1500	300	...	Liquid 40
40	Ring	130.101	1200	...	...	Ring R1 Fr40
50	Sadl	626.268	2190	1619	...	Left Saddle
50	Liqd	22493	1450	300	...	Liquid 50
50	Ring	130.265	600	...	...	Ring R1 Fr50
50	Nozl	42.4279	2650	1544.45	...	N4-3''
50	Nozl	88.449	2050	1584.14	...	N5-6
50	Nozl	669.162	1300	1804.8	...	M2-24''
50	Nozl	27.8881	1450	-1557.15	...	N8-4''
50	Wght	50	1300	1650	...	DAVIT-2
50	Wght	85.0015	2050	1200	...	DIP pipe
50	Wght	220	1400	1212.44	700	LL-2
60	Liqd	4456.82	300	300	...	Liquid 60

Total Weight of Each Detail Type:

Saddles	1252.5
Liquid	101212.7
Stiffeners	518.9
Nozzles	2027.3
Weights	753.0

Sum of the Detail Weights 105764.4 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	12571.6	12571.6	12571.6	12571.6	12571.6	12571.6
Saddles	1252.5	1252.5	1252.5	1252.5	1252.5	1252.5
Stif. Rings	518.9	518.9	518.9	518.9	518.9	518.9
Nozzles	2027.3	2027.3	2027.3	2027.3	2027.3	2027.3
Wld Weights	540.0	540.0	540.0	540.0	540.0	540.0
Empty Weights	...	...	...	213.0	213.0	...
Test Weights	...	213.0	...	...	...	...
Ope Weights	...	...	...	...	...	213.0
Ope. Liquid	...	...	...	...	...	101212.7
Test Liquid	...	91852.0	...	...	...	...
Totals	16910.3	108975.3	16910.3	17123.3	17123.3	118336.0

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	16910.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	108975.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	16910.3 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	17123.3 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	17123.3 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	118336.0 kgm
Oper. Wt. + CA	- Corr wt. + operating liquid	118336.0 kgm
Field Test Wt.	- Empty weight + test medium (full)	108975.3 kgm

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	10360759	10227107
20	30	28500530	28274334
30	40	28462830	28274334
40	50	28462830	28274334
50	60	27550514	27331858
60	70	10360759	10227107

Total	133698208.0	132609088.0 mm ²
Total	133.7	132.6 m ²

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**Nozzle Flange MAWP Results: (MPa & °C)**

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-6''	3.93	4.97	120	300	GR 2.1	3.508	2.384	3.928	2.384
N9-6''	3.93	4.97	120	300	GR 2.1	...	...	...	...
N2-3''	3.93	4.97	120	300	GR 2.1	4.878	1.015	3.928	1.015
N3-6''	3.93	4.97	120	300	GR 2.1	3.509	2.384	3.928	2.384
M1-24''	1.50	1.91	120	150	GR 2.1	...	...	...	...
N7-4''	3.93	4.97	120	300	GR 2.1	4.480	1.413	3.928	1.413
N6-4''	3.93	4.97	120	300	GR 2.1	4.480	1.413	3.928	1.413
K1-2''	1.50	1.91	120	150	GR 2.1	...	...	...	...
K2-4''	1.50	1.91	120	150	GR 2.1	...	...	...	...
K3-2''	1.50	1.91	120	150	GR 2.1	...	...	...	...
N4-3''	3.93	4.97	120	300	GR 2.1	4.878	1.015	3.928	1.015
N5-6	3.93	4.97	120	300	GR 2.1	3.509	2.384	3.928	2.384
M2-24''	1.50	1.91	120	150	GR 2.1	...	...	...	...
N8-4''	3.93	4.97	120	300	GR 2.1	4.480	1.413	3.928	1.413
Min Rating	1.015	1.910, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1-6''	196.088	0.50	4724851	9600	0.547	2.384	Pass
N2-3''	111.125	0.50	1181304	4800	0.547	1.015	Pass
N3-6''	196.088	0.50	4725218	9600	0.547	2.384	Pass
N7-4''	138.176	0.50	2100096	6400	0.547	1.413	Pass
N6-4''	138.176	0.50	2100096	6400	0.547	1.413	Pass
N4-3''	111.125	0.50	1181304	4800	0.547	1.015	Pass
N5-6	196.088	0.50	4725218	9600	0.547	2.384	Pass
N8-4''	138.176	0.50	2100096	6400	0.547	1.413	Pass

Evaluating Flange on Nozzle [N1-6'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4724851) + 4 \cdot \text{abs}(9600) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((3.928 - 0.547) + .500 \cdot 3.928)$$

$$= 0.83130E+08 \leq 0.12666E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-6'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 83130112.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.928, 2.384)$$

$$= 2.384 \text{ MPa}$$

Evaluating Flange on Nozzle [N2-3'']:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800) \cdot 111.125 \leq \pi \cdot 111.125^3 \cdot ((3.928 - 0.547) + .500 \cdot 3.928)$$

$$= 0.21035E+08 \leq 0.23054E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-3'' [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 21035196.0 / (\pi \cdot 111.125^3))$$



$$= \min(3.928, 1.015)$$

$$= 1.015 \text{ MPa}$$

Evaluating Flange on Nozzle [N3-6"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(4725218) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.83136E+08 \leq 0.12666E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-6" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 83135992.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.928, 2.384)$$

$$= 2.384 \text{ MPa}$$

Evaluating Flange on Nozzle [N7-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.37140E+08 \leq 0.44320E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N7-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(3.928, 1.413)$$

$$= 1.413 \text{ MPa}$$

Evaluating Flange on Nozzle [N6-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.37140E+08 \leq 0.44320E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(3.928, 1.413)$$

$$= 1.413 \text{ MPa}$$

Evaluating Flange on Nozzle [N4-3"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1181304) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.21035E+08 \leq 0.23054E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-3" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 21035196.0 / (\pi \cdot 111.125^3))$$

$$= \min(3.928, 1.015)$$

$$= 1.015 \text{ MPa}$$

Evaluating Flange on Nozzle [N5-6]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(4725218) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.83136E+08 \leq 0.12666E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N5-6 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 83135992.0 / (\pi \cdot 196.088^3))$$

$$= \min(3.928, 2.384)$$

$$= 2.384 \text{ MPa}$$

Evaluating Flange on Nozzle [N8-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2100096) + 4 \cdot \text{abs}(6400)138.176 \leq \pi \cdot 138.176^3 [(3.928 - 0.547) + .500 \cdot 3.928]$$

$$= 0.37140E+08 \leq 0.44320E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N8-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.928, 0.500 \cdot 3.928 + 3.928 - 37140128.0 / (\pi \cdot 138.176^3))$$

$$= \min(3.928, 1.413)$$

$$= 1.413 \text{ MPa}$$



Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * S_{ms} = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * S_{m1} = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.70.75)$
 = 0.118 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.12$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot S_{ds} \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 1148119 \cdot 1.5$
 = 112286.133 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot S_{ds} \cdot W$
 = $0.2 \cdot 0.22 \cdot 1148119$
 = 49904.9 N

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N
10	20	1500	143515	14035.8
20	30	1500	143515	14035.8
20	30	1500	143515	14035.8
30	40	1500	143515	14035.8
40	50	1500	143515	14035.8



50	0	1500	143515	14035.8
50	60	1500	143515	14035.8
60	70	1500	143515	14035.8

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5985.000 mm
Center of Gravity of the Liquid	6000.000 mm
Center of Gravity of the Stiffening Rings	6050.074 mm
Center of Gravity of the Nozzles	5286.305 mm
Center of Gravity of the Added Weights (Operating)	5798.060 mm
Center of Gravity of the Added Weights (Empty)	5798.060 mm
Center of Gravity of Bare Shell New and Cold	5996.110 mm
Center of Gravity of Bare Shell Corroded	5996.110 mm
Vessel CG in the Operating Condition	5986.136 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	5904.188 mm
Vessel CG in the Test Condition	5984.944 mm

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**ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:**

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

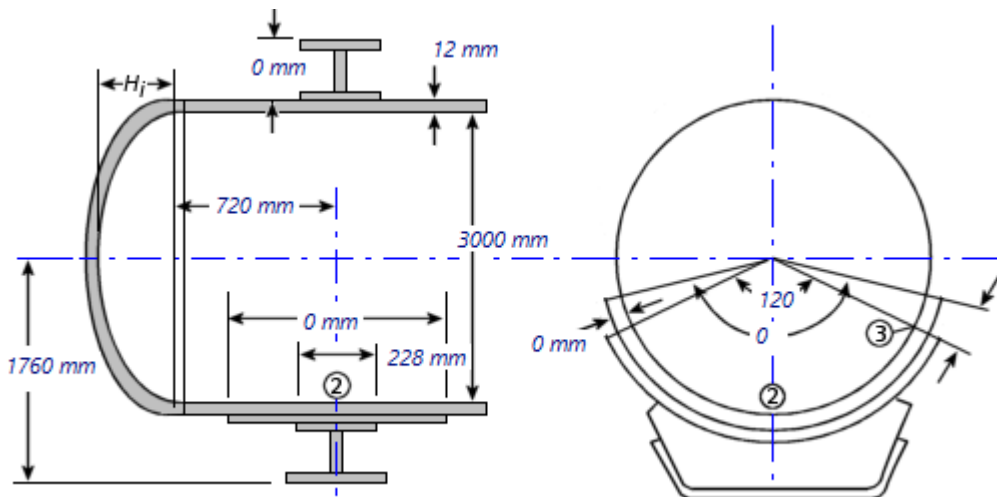
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{1506 \cdot 12}, 2 \cdot 720)$$

$$= 437.7142 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm
Shell Allowable Stress used in Calculation		108.20	MPa
Head Allowable Stress used in Calculation		108.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

**Results for Horizontal Vessel Analysis:**

Saddle Force Q, Operating Case	711335.88	N
Pressure used in Analysis	P	0.361 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	6.10	108.20
Long. Stress at Bottom of Midspan	48.31	108.20
Long. Stress at Top of Saddles	22.92	108.20
Long. Stress at Bottom of Saddles	22.42	108.20
Tangential Shear in Shell	34.64	86.56



Tangential Shear in Head	51.96	86.56
Circ. Stress at Horn of Saddle	132.16	135.25
Addl. Stress in Head as Stiffener	91.06	135.25
Circ. Compressive Stress in Shell	10.30	108.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(FI, eq; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(112286, 0) \cdot 1760/10510$$

$$= 18803.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (112286/2 + 0) \cdot 1760/2636$$

$$= 112448.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 598887 + \max(0, 0, 18803, 112449)$$

$$= 711335.9 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	717477.06	N
Transverse Shear Load	56143.07	N
Longitudinal Shear Load	112286.12	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3000/2 + 12 + 10)\sin(120/2), 2 \cdot 1506)$$

$$= 2636.181 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -711336 \cdot 720 [1 - (1 - 720/12000 + (1506^2 - 754^2)/(2 \cdot 720 \cdot 12000))/(1 + (4 \cdot 754)/(3 \cdot 12000))]$$

$$= -21475662.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 711336 \cdot 12000/4 ((1 + 2(1506^2 - 754^2)/(12000^2))/(1 + (4 \cdot 754)/(3 \cdot 12000)) - 4 \cdot 720/12000)$$

$$= 1503971456.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.36 \cdot 1506/(2 \cdot 10) - 1503971456/(\pi \cdot 1506^2 \cdot 10)$$

$$= 6.10 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.36 \cdot 1506/(2 \cdot 10) + 1503971456/(\pi \cdot 1506^2 \cdot 10)$$

$$= 48.31 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.36 \cdot 1506/(2 \cdot 12) - 21475662/(\pi \cdot 1506^2 \cdot 12)$$

$$= 22.92 \text{ MPa}$$



Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 0.36 \cdot 1506 / (2 \cdot 12) + -21475662 / (\pi \cdot 1506^2 \cdot 12)$$

$$= 22.42 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 711336 (12000 - 2 \cdot 720) / (12000 + (4 \cdot 754 / 3))$$

$$= 577586.7 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (R_m \cdot t)$$

$$= 0.88 \cdot 711336 / (1506 \cdot 12)$$

$$= 34.64 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.88 \cdot 711336 / (1506 \cdot 8)$$

$$= 51.96 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4 \cdot 711336 / (1506 \cdot 8) + ((0.36 \cdot 1500) / (2 \cdot 8)) \cdot (1500 / 754)$$

$$= 91.06 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 711336 \cdot 0.1 / (12(228 + 105 + 105))$$

$$= -10.30 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -711336 / (4 \cdot 12(228 + 105 + 105)) - 12 \cdot 0.013 \cdot 711336 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -132.16 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 10510.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.16364\text{E-}04 \cdot 10510 (120 - 21.1)$$

$$= 17.007 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	216.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280)^3) / 12$$

$$= 70458.555 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 70459 + 5872 (123 - 6)^2$$

$$= 80877888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478059 / 20096$$

$$= 123.314 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 711336$$

$$= 144772.2812 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.1789 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1760 - 1500 \cdot \sin(120 / 2) / 1.05$$

$$= 519.510 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 75241128.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 38.1241 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(711336 + 6141)254 / (4 \cdot 2668 \cdot 138)}$$

$$= 19.268 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120 / 2) (1500 + 12 + 10)$$

$$= 2636.181 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2636 / (4 - 1)$$

$$= 878.727 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 879 \cdot 254 / 2$$

$$= 111598.336 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 717477 / (2668 \cdot 254)$$

$$= 1.059 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 111598 \cdot 1.06$$

$$= 118153.375 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 2756 + 7030$$

$$= 9785.816 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 118153 / 9786$$

$$= 12.075 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 228 - 16$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 9786$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 228 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (12840037 / 9786) }$$

$$= 36.223 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1760 - \cos(120 / 2) (1500 + 12 + 10) - 32$$

$$= 967.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 202713 / 207) }$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 967 / 36.2$$

$$= 26.696$$



Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 112286 / (2 \cdot 2668) \cdot 879 \cdot 967 / 2$$

$$= 8944102.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (26.7 < 139) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.7)^2 / (2 \cdot 139^2)) 207 /$$

$$(5/3 + 3 \cdot (26.7) / (8 \cdot 139) - (26.7^3) / (8 \cdot 139^3))$$

$$= 116.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.1 / 117 + (8944102 \cdot 114 / 12840049) / 138$$

$$= 0.679$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 16816$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1760 - \sqrt{ (1522 + 10)^2 - (879/2)^2 } - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 13135538 / 16816 }$$

$$= 27.949 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 271 / 27.9$$

$$= 9.689$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 112286 / (2 \cdot 2668)$$

$$= 21.043 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 21 \cdot 879 \cdot 271$$

$$= 5009361.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5009361 / 115224$$

$$= 43.461 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (9.69 < 139) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 139^2)) 207 /$$

$$(5/3 + 3 \cdot (9.69) / (8 \cdot 139) - (9.69^3) / (8 \cdot 139^3))$$



= 122 MPa

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $14/122 + (5009361 \cdot 114/13135538) / 138$
 = 0.430

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	84.0 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	707.1440 mm ²
Saddle Load QO (Weight)	QO	605028.3 N
Saddle Load QL (Wind/Seismic contribution)	QL	18803.4 N
Maximum Transverse Force	Ft	56143.1 N
Maximum Longitudinal Force	Fl	112286.1 N
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 363167.3 N

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $\max(Ff, Fl) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $363167 / (707 \cdot 8)$
 = 64.2 MPa. Must be less than 100.0 MPa.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $56143 / (707 \cdot 8)$
 = 9.9 MPa. Must be less than 100.0 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $112286 / (100 \cdot 8)$
 = 140.3696 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1760 \cdot 56143 + 0$
 = 98851864.00 N-mm

Eccentricity (e):
 = $Rmom / QO$
 = $98851864 / 605028$
 = 163.32 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 140, 0]$
 = 140.3696 mm²

**ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:****Note:**

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(228 + 1.56 \sqrt{ (1506 \cdot 12) }, 2 \cdot 720)$$

$$= 437.7142 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm
Shell Allowable Stress used in Calculation		108.20	MPa
Head Allowable Stress used in Calculation		108.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		711586.50	N
Pressure used in Analysis	P	0.361	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	6.10	108.20
Long. Stress at Bottom of Midspan	48.31	108.20
Long. Stress at Top of Saddles	22.92	108.20
Long. Stress at Bottom of Saddles	22.42	108.20
Tangential Shear in Shell	34.65	86.56
Tangential Shear in Head	51.97	86.56
Circ. Stress at Horn of Saddle	132.21	135.25
Addl. Stress in Head as Stiffener	91.07	135.25
Circ. Compressive Stress in Shell	10.30	108.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl_{eq}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(112286, 0) \cdot 1760 / 10510$$

$$= 18803.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr} (Ft / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (112286 / 2 + 0) \cdot 1760 / 2636$$

$$= 112448.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 599138 + \max(0, 0, 18803, 112449)$$

$$= 711586.5 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

Vertical Load (including saddle weight)	717727.69	N
Transverse Shear Load	56143.07	N
Longitudinal Shear Load	112286.12	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(3000/2 + 12 + 10)\sin(120/2), 2 \cdot 1506)$$

$$= 2636.181 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -711586 \cdot 720 [1 - (1 - 720/12000 + (1506^2 - 754^2)/(2 \cdot 720 \cdot 12000))/(1 + (4 \cdot 754)/(3 \cdot 12000))]$$

$$= -21483228.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 711586 \cdot 12000/4((1 + 2(1506^2 - 754^2)/(12000^2))/(1 + (4 \cdot 754)/(3 \cdot 12000)) - 4 \cdot 720/12000)$$

$$= 1504501376.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.36 \cdot 1506/(2 \cdot 10) - 1504501376/(\pi \cdot 1506^2 \cdot 10)$$

$$= 6.10 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.36 \cdot 1506/(2 \cdot 10) + 1504501376/(\pi \cdot 1506^2 \cdot 10)$$

$$= 48.31 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.36 \cdot 1506/(2 \cdot 12) - 21483228/(\pi \cdot 1506^2 \cdot 12)$$

$$= 22.92 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 t)$$

$$= 0.36 \cdot 1506/(2 \cdot 12) + 21483228/(\pi \cdot 1506^2 \cdot 12)$$

$$= 22.42 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 711586(12000 - 2 \cdot 720)/(12000 + (4 \cdot 754/3))$$

$$= 577790.1 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (R_m \cdot t)$$

$$= 0.88 \cdot 711586 / (1506 \cdot 12)$$

$$= 34.65 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.88 \cdot 711586 / (1506 \cdot 8)$$

$$= 51.97 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K4 \cdot Q/(R_m \cdot t_h) + ((P \cdot R_i)/(2 \cdot t_h)) \cdot (R_i/h_m)$$

$$= 0.4 \cdot 711586/(1506 \cdot 8) +$$



$$((0.36 \cdot 1500)/(2 \cdot 8)) \cdot (1500/754)$$

$$= 91.07 \text{ MPa}$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 711586 \cdot 0.1 / (12(228 + 105 + 105))$$

$$= -10.30 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -711586 / (4 \cdot 12(228 + 105 + 105)) -$$

$$12 \cdot 0.013 \cdot 711586 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -132.21 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	216.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280) 12^3) / 12$$

$$= 70458.555 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 70459 + 5872(123 - 6)^2$$

$$= 80877888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478059/20096$$

$$= 123.314 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$



Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 711586$$

$$= 144823.2812 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.1825 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1500 \cdot \sin(120/2) / 1.05$$

$$= 519.510 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 75267640.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 38.1375 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(711586 + 6141) \cdot 254) / (4 \cdot 2668 \cdot 138)}$$

$$= 19.272 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1500 + 12 + 10)$$

$$= 2636.181 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2636 / (4 - 1)$$

$$= 878.727 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 879 \cdot 254 / 2$$

$$= 111598.336 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 717728 / (2668 \cdot 254)$$

$$= 1.059 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 111598 \cdot 1.06$$

$$= 118194.664 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2756 + 7030$$

$$= 9785.816 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 118195 / 9786$$

$$= 12.079 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

= Saddle Width - Web Thickness

= 228 - 16

= 212.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/9786

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 228/2

= 114.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(12840037/9786)}$

= 36.223 mm

Length of Outer Rib [L]:

= Saddle Height - $\cos(\theta/2)$ (radius + shlthk + wpdthk) - bpthk

= 1760 - $\cos(120/2)$ (1500 + 12 + 10) - 32

= 967.000 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 202713/207)}$

= 139.035

Slenderness ratio [KL/r]:

= KL/r

= 1 • 967/36.2

= 26.696

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 112286 / (2 • 2668) • 879 • 967/2

= 8944102.000 N-mm

Compressive Allowable, $KL/r < Cc$ (26.7 < 139) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (26.7)^2 / (2 \cdot 139^2)) 207 /$

$(5/3 + 3 \cdot (26.7) / (8 \cdot 139) - (26.7)^3 / (8 \cdot 139^3))$

= 116.9 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 12.1/117 + (8944102 • 114/12840049) / 138

= 0.679

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/16816

= 0.000 mm



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228 / 2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1760 - \sqrt{(1522 + 10)^2 - (879/2)^2} - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13135538 / 16816)}$$

$$= 27.949 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 271 / 27.9$$

$$= 9.689$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 112286 / (2 \cdot 2668)$$

$$= 21.043 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 21 \cdot 879 \cdot 271$$

$$= 5009361.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5009361 / 115224$$

$$= 43.461 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.69 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 139^2)) 207 /$$

$$(5/3 + 3 \cdot (9.69) / (8 \cdot 139) - (9.69^3) / (8 \cdot 139^3))$$

$$= 122 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 14.1/122 + (5009361 \cdot 114 / 13135538) / 138$$

$$= 0.430$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	84.0 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BlArea	707.1440 mm ²
Saddle Load Q0 (Weight)	Q0	605278.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	18803.4 N
Maximum Transverse Force	Ft	56143.1 N
Maximum Longitudinal Force	Fl	112286.1 N
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 112286 / (707 \cdot 8)$$

$$= 19.9 \text{ MPa. Must be less than } 100.0 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $56143 / (707 \cdot 8)$
= 9.9 MPa. Must be less than 100.0 MPa.

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_t / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $112286 / (100 \cdot 8)$
= 140.3696 mm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1760 \cdot 56143 + 0$
= 98851864.00 N-mm

Eccentricity (e):
= R_{mom} / QO
= $98851864 / 605279$
= 163.25 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 140, 0]$
= 140.3696 mm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

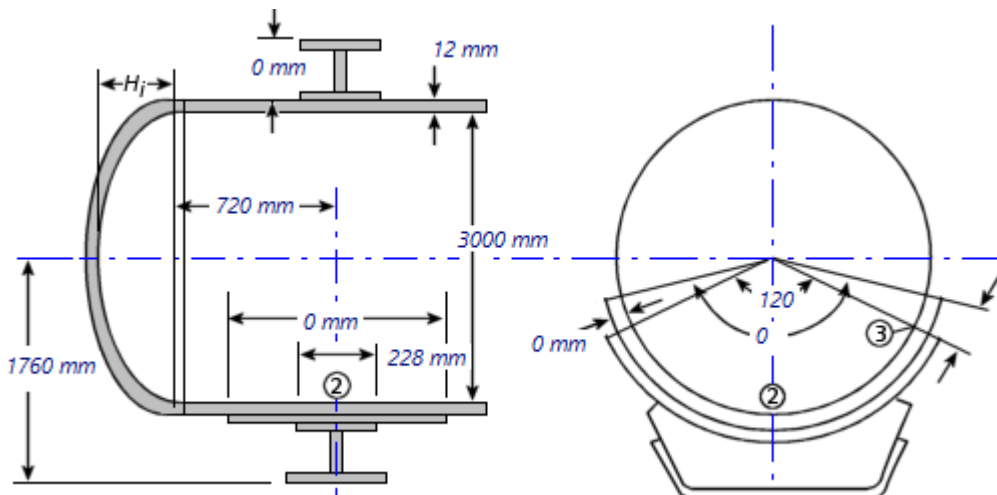
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{1506 \cdot 12}, 2 \cdot 720)$$

$$= 437.7142 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm
Shell Allowable Stress used in Calculation		184.50	MPa
Head Allowable Stress used in Calculation		184.50	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	528169.88	N
Pressure used in Analysis	P	0.725 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	38.96	184.50
Long. Stress at Bottom of Midspan	70.29	184.50
Long. Stress at Top of Saddles	45.71	184.50
Long. Stress at Bottom of Saddles	45.33	184.50
Tangential Shear in Shell	25.72	109.26



Tangential Shear in Head	38.58	86.77
Circ. Stress at Horn of Saddle	98.13	276.74
Addl. Stress in Head as Stiffener	152.88	184.50
Circ. Compressive Stress in Shell	7.65	184.50

There were no external loads for this load condition.

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 528170 + max(0, 0, 0, 0)

= 528169.9 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 534311.12 N

Transverse Shear Load 0.00 N

Longitudinal Shear Load 0.00 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(3000/2 + 12 + 10) \sin(120/2), 2 \cdot 1506)$

= 2636.181 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0132 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$

= $-528170 \cdot 720 [1 - (1 - 720/12000 + (1506^2 - 754^2)/$

$(2 \cdot 720 \cdot 12000) / (1 + (4 \cdot 754) / (3 \cdot 12000))]$

= -15945771.0 N-mm

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$

= $528170 \cdot 12000 / 4 ((1 + 2(1506^2 - 754^2)/(12000^2)) / (1 + (4 \cdot 754)/$

$(3 \cdot 12000)) - 4 \cdot 720 / 12000)$

= 1116705152.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t)$

= $0.73 \cdot 1506 / (2 \cdot 10) - 1116705152 / (\pi \cdot 1506^2 \cdot 10)$

= 38.96 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$

= $0.73 \cdot 1506 / (2 \cdot 10) + 1116705152 / (\pi \cdot 1506^2 \cdot 10)$

= 70.29 MPa

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

= $P \cdot R_m / (2t) - M1 / (\pi \cdot R_m^2 \cdot t)$

= $0.73 \cdot 1506 / (2 \cdot 12) - 15945771 / (\pi \cdot 1506^2 \cdot 12)$

= 45.71 MPa

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

= $P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 \cdot t)$

= $0.73 \cdot 1506 / (2 \cdot 12) + 15945771 / (\pi \cdot 1506^2 \cdot 12)$

= 45.33 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a) / (L + (4 \cdot h m / 3))$

= $528170 (12000 - 2 \cdot 720) / (12000 + (4 \cdot 754 / 3))$

= 428860.6 N



Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.88 \cdot 528170 / (1506 \cdot 12)$$

$$= 25.72 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.88 \cdot 528170 / (1506 \cdot 8)$$

$$= 38.58 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4 \cdot 528170 / (1506 \cdot 8) +$$

$$((0.73 \cdot 1500) / (2 \cdot 8)) \cdot (1500 / 754)$$

$$= 152.88 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t (b + X_1 + X_2))$$

$$= -0.76 \cdot 528170 \cdot 0.1 / (12 (228 + 105 + 105))$$

$$= -7.65 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t (b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -528170 / (4 \cdot 12 (228 + 105 + 105)) -$$

$$12 \cdot 0.013 \cdot 528170 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -98.13 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	205.0	MPa
Height of Web at Center	Hw,c	216.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280)^3) / 12$$

$$= 70458.555 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 70459 + 5872 (123 - 6)^2$$

$$= 80877888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09



Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478059/20096$$

$$= 123.314 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 528170$$

$$= 107494.0234 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 7.5579 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 122.9978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1500 \cdot \sin(120/2) / 1.05$$

$$= 519.510 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 55866856.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 28.3073 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 136.6642 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(528170 + 6141)254 / (4 \cdot 2668 \cdot 137)}$$

$$= 16.709 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1500 + 12 + 10)$$

$$= 2636.181 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2636 / (4 - 1)$$

$$= 878.727 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 879 \cdot 254/2$$

$$= 111598.336 \text{ mm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 534311 / (2668 \cdot 254)$$

$$= 0.788 \text{ N/mm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 111598 \cdot 0.79$$

$$= 87989.805 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2756 + 7030$$



$$= 9785.816 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 87990/9786$$

$$= 8.992 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228 - 16$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/9786$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228/2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(12840037/9786)}$$

$$= 36.223 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(120/2) (1500 + 12 + 10) - 32$$

$$= 967.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713/205)}$$

$$= 139.712$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 967/36.2$$

$$= 26.696$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 0 / (2 \cdot 2668) \cdot 879 \cdot 967/2$$

$$= 0.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ ($26.7 < 140$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.7)^2 / (2 \cdot 140^2)) 205 /$$

$$(5/3 + 3 \cdot (26.7) / (8 \cdot 140) - (26.7^3) / (8 \cdot 140^3))$$

$$= 115.8 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 8.99/116 + (0 \cdot 114/12840049) / 137$$

$$= 0.078$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction



	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/16816$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1760 - \sqrt{(1522 + 10)^2 - (879/2)^2} - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13135538/16816)}$$

$$= 27.949 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 271/27.9$$

$$= 9.689$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 0/(2 \cdot 2668)$$

$$= 0.000 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0 \cdot 879 \cdot 271$$

$$= 0.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0/115224$$

$$= 0.000 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.69 < 140$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 140^2)) 205 /$$

$$(5/3 + 3 \cdot (9.69) / (8 \cdot 140) - (9.69)^3 / (8 \cdot 140^3))$$

$$= 120.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 10.5/121 + (0 \cdot 114/13135538) / 137$$

$$= 0.087$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	84.0 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	707.1440 mm ²



Saddle Load Q0 (Weight)	Q0	534311.1 N
Saddle Load QL (Wind/Seismic contribution)	QL	0.0 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	Fl	0.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0 / (707 \cdot 8)$
 $= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0 / (707 \cdot 8)$
 $= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 0 / (68.4 \cdot 8)$
 $= 0.0000 \text{ mm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1760 \cdot 0 + 0$
 $= 0.00 \text{ N-mm}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 0 / 534311$
 $= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 0, 0]$
 $= 0.0000 \text{ mm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 $= \min(228 + 1.56 \cdot \sqrt{1506 \cdot 12}, 2 \cdot 720)$
 $= 437.7142 \text{ mm}$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm



Shell Allowable Stress used in Calculation	184.50	MPa
Head Allowable Stress used in Calculation	184.50	MPa
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	

Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	528158.69	N
Pressure used in Analysis	P	0.725 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	38.96	184.50
Long. Stress at Bottom of Midspan	70.29	184.50
Long. Stress at Top of Saddles	45.71	184.50
Long. Stress at Bottom of Saddles	45.33	184.50
Tangential Shear in Shell	25.72	109.26
Tangential Shear in Head	38.58	86.77
Circ. Stress at Horn of Saddle	98.13	276.74
Addl. Stress in Head as Stiffener	152.88	184.50
Circ. Compressive Stress in Shell	7.65	184.50

There were no external loads for this load condition.

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(FwL, FwT, Fsl, Fst)

= 528159 + max(0, 0, 0, 0)

= 528158.7 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	534299.88	N
Transverse Shear Load	0.00	N
Longitudinal Shear Load	0.00	N

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(3000/2 + 12 + 10)sin(120/2), 2•1506)

= 2636.181 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]= -528159•720[1-(1-720/12000+(1506²-754²)/

(2•720•12000))/(1+(4•754)/(3•12000))]

= -15945433.0 N-mm

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)= 528159•12000/4((1+2(1506²-754²)/(12000²))/(1+(4•754)/

(3•12000))-4•720/12000)

= 1116681472.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)



$$= 0.73 \cdot 1506 / (2 \cdot 10) - 1116681472 / (\pi \cdot 1506^2 \cdot 10)$$

$$= 38.96 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.73 \cdot 1506 / (2 \cdot 10) + 1116681472 / (\pi \cdot 1506^2 \cdot 10)$$

$$= 70.29 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.73 \cdot 1506 / (2 \cdot 12) - 15945433 / (\pi \cdot 1506^2 \cdot 12)$$

$$= 45.71 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.73 \cdot 1506 / (2 \cdot 12) + 15945433 / (\pi \cdot 1506^2 \cdot 12)$$

$$= 45.33 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 528159 (12000 - 2 \cdot 720) / (12000 + (4 \cdot 754 / 3))$$

$$= 428851.4 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.88 \cdot 528159 / (1506 \cdot 12)$$

$$= 25.72 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.88 \cdot 528159 / (1506 \cdot 8)$$

$$= 38.58 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4 \cdot 528159 / (1506 \cdot 8) + ((0.73 \cdot 1500) / (2 \cdot 8)) \cdot (1500 / 754)$$

$$= 152.88 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 528159 \cdot 0.1 / (12(228 + 105 + 105))$$

$$= -7.65 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -528159 / (4 \cdot 12(228 + 105 + 105)) - 12 \cdot 0.013 \cdot 528159 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -98.13 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	

Saddle Yield Stress Sy 205.0 MPa
 Height of Web at Center Hw,c 216.0 mm
 Friction Coefficient mu 0.600

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280)^3) / 12$
 $= 70458.555 \text{ mm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 70459 + 5872 (123 - 6)^2$
 $= 80877888.000 \text{ mm}^4$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ⁽²⁾ mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09

Distance to Centroid [C1]:
 $= AY / A$
 $= 2478059 / 20096$
 $= 123.314 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 528159$
 $= 107491.7500 \text{ N}$

Tension Stress, St = (Fh/As) = 7.5577 MPa
 Allowed Stress, Sa = 0.6 * Yield Str = 122.9978 MPa

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1760 - 1500 \cdot \sin(120/2) / 1.05$
 $= 519.510 \text{ mm}$

Bending Moment, M = Fh * d = 55865676.0000 N-mm

Bending Stress, Sb = (M * C1 / I) = 28.3067 MPa
 Allowed Stress, Sa = 2/3 * Yield Str = 136.6642 MPa

Minimum Thickness of Baseplate per Moss:
 $= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(528159 + 6141)254) / (4 \cdot 2668 \cdot 137)}$
 $= 16.709 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1500 + 12 + 10)$
 $= 2636.181 \text{ mm}$

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2636 / (4 - 1)
 = 878.727 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 879 • 254 / 2
 = 111598.336 mm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 534300 / (2668 • 254)
 = 0.788 N/mm²

Axial Load [P]:
 = Ap • Bp
 = 111598 • 0.79
 = 87987.953 N

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 2756 + 7030
 = 9785.816 mm²

Compressive Stress [Sc]:
 = P / Ar
 = 87988 / 9786
 = 8.992 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 228 - 16
 = 212.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0 / 9786
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 228 / 2
 = 114.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{12840037 / 9786}$
 = 36.223 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1760 - cos(120/2)(1500 + 12 + 10) - 32
 = 967.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 202713 / 205}$
 = 139.712

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 967/36.2$$

$$= 26.696$$

Bending Moment $[Rm]$:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 0 / (2 \cdot 2668) \cdot 879 \cdot 967/2$$

$$= 0.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < C_c$ (26.7 < 140) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.7)^2 / (2 \cdot 140^2)) 205 /$$

$$(5/3 + 3 \cdot (26.7) / (8 \cdot 140) - (26.7)^3 / (8 \cdot 140^3))$$

$$= 115.8 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 8.99/116 + (0 \cdot 114/12840049) / 137$$

$$= 0.078$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum $[y_{tot}]$:

$$= AY / A$$

$$= 0/16816$$

$$= 0.000 \text{ mm}$$

Distance to Centroid $[C1]$:

$$= \text{Saddle Width} / 2$$

$$= 228/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib $[L]$:

$$= \text{Saddle Height} - \sqrt{(Ro + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 1760 - \sqrt{(1522 + 10)^2 - (879/2)^2} - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration $[r]$:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{13135538/16816}$$

$$= 27.949 \text{ mm}$$

Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 271/27.9$$

$$= 9.689$$

Unit Force $[Force, u]$:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 0 / (2 \cdot 2668)$$

$$= 0.000 \text{ N/mm}$$

Moment at base of inner Rib $[M_{base, c}]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0 \cdot 879 \cdot 271$$

$$= 0.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma_{B, base, c}]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 0/115224$$

$$= 0.000 \text{ MPa}$$



Compressive Allowable, $KL/r < C_c$ ($9.69 < 140$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 140^2)) 205 /$$

$$(5/3 + 3 \cdot (9.69) / (8 \cdot 140) - (9.69^3) / (8 \cdot 140^3))$$

$$= 120.8 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 10.5 / 121 + (0 \cdot 114 / 13135538) / 137$$

$$= 0.087$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 MPa
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	84.0 mm
Nominal Bolt Diameter	Bnd	42.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	108.00 MPa
Area Available in a Single Bolt	BltArea	707.1440 mm ²
Saddle Load Q0 (Weight)	Q0	534299.9 N
Saddle Load QL (Wind/Seismic contribution)	QL	0.0 N
Maximum Transverse Force	Ft	0.0 N
Maximum Longitudinal Force	Fl	0.0 N
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (707 \cdot 8)$$

$$= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (707 \cdot 8)$$

$$= 0.0 \text{ MPa. Must be less than } 68.4 \text{ MPa.}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 0 / (68.4 \cdot 8)$$

$$= 0.0000 \text{ mm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1760 \cdot 0 + 0$$

$$= 0.00 \text{ N-mm}$$

Eccentricity (e):

$$= R_{mom} / Q_0$$

$$= 0 / 534300$$

$$= 0.00 \text{ mm} < B_{plen} / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0, 0]$$

$$= 0.0000 \text{ mm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

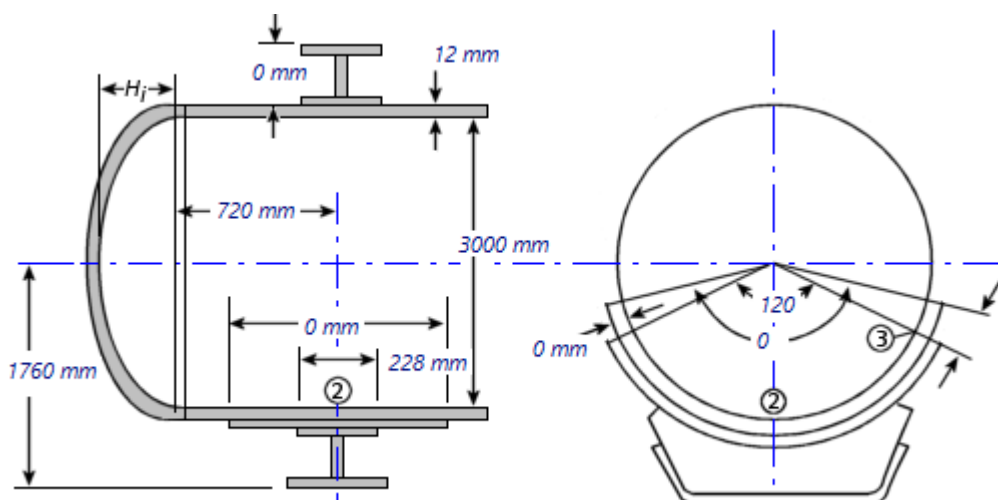
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{1506 \cdot 12}, 2 \cdot 720)$$

$$= 437.7142 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm
Shell Allowable Stress used in Calculation		108.20	MPa
Head Allowable Stress used in Calculation		108.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	711335.88	N
Pressure used in Analysis	P	-0.100 MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-28.63	-93.67
Long. Stress at Top of Midspan	28.63	108.20
Long. Stress at Bottom of Midspan	13.57	108.20
Long. Stress at Top of Saddles	-6.02	-112.33
Long. Stress at Top of Saddles	6.02	108.20
Long. Stress at Bottom of Saddles	-6.53	-112.33



Long. Stress at Bottom of Saddles	6.53	108.20
Tangential Shear in Shell	34.64	86.56
Tangential Shear in Head	51.96	86.56
Circ. Stress at Horn of Saddle	132.16	135.25
Addl. Stress in Head as Stiffener	5.03	135.25
Circ. Compressive Stress in Shell	10.30	108.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(FI, eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(112286, 0) \cdot 1760/10510$$

$$= 18803.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3(112286/2 + 0) \cdot 1760/2636$$

$$= 112448.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 598887 + \max(0, 0, 18803, 112449)$$

$$= 711335.9 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 717477.06 N

Transverse Shear Load 56143.07 N

Longitudinal Shear Load 112286.12 N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(3000/2 + 12 + 10)\sin(120/2), 2 \cdot 1506)$$

$$= 2636.181 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0132 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is less than $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -711336 \cdot 720 [1 - (1 - 720/12000 + (1506^2 - 754^2)/$$

$$(2 \cdot 720 \cdot 12000)/(1 + (4 \cdot 754)/(3 \cdot 12000))]$$

$$= -21475662.0 \text{ N-mm}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 711336 \cdot 12000/4((1 + 2(1506^2 - 754^2)/(12000^2))/(1 + (4 \cdot 754)/$$

$$(3 \cdot 12000)) - 4 \cdot 720/12000)$$

$$= 1503971456.0 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1506/(2 \cdot 10) - 1503971456/(\pi \cdot 1506^2 \cdot 10)$$

$$= -28.63 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= -0.1 \cdot 1506/(2 \cdot 10) + 1503971456/(\pi \cdot 1506^2 \cdot 10)$$

$$= 13.57 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$



$$= -0.1 \cdot 1506 / (2 \cdot 12) - 21475662 / (\pi \cdot 1506^2 \cdot 12)$$

$$= -6.02 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= -0.1 \cdot 1506 / (2 \cdot 12) + -21475662 / (\pi \cdot 1506^2 \cdot 12)$$

$$= -6.53 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 711336 \cdot (12000 - 2 \cdot 720) / (12000 + (4 \cdot 754 / 3))$$

$$= 577586.7 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.88 \cdot 711336 / (1506 \cdot 12)$$

$$= 34.64 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.88 \cdot 711336 / (1506 \cdot 8)$$

$$= 51.96 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4 \cdot 711336 / (1506 \cdot 8) + ((-0.1 \cdot 1500) / (2 \cdot 8)) \cdot (1500 / 754)$$

$$= 5.03 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 711336 \cdot 0.1 / (12(228 + 105 + 105))$$

$$= -10.30 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{7*}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -711336 / (4 \cdot 12(228 + 105 + 105)) - 12 \cdot 0.013 \cdot 711336 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -132.16 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	216.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280)^3) / 12$$

$$= 70458.555 \text{ mm}^4$$



Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 70459 + 5872 (123 - 6)^2$$

$$= 80877888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478059/20096$$

$$= 123.314 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 711336$$

$$= 144772.2812 \text{ N}$$

$$\text{Tension Stress, } St = (Fh / As) = 10.1789 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1500 \cdot \sin(120/2) / 1.05$$

$$= 519.510 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 75241128.0000 \text{ N-mm}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 38.1241 \text{ MPa}$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(711336 + 6141)254) / (4 \cdot 2668 \cdot 138)}$$

$$= 19.268 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1500 + 12 + 10)$$

$$= 2636.181 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2636 / (4 - 1)$$

$$= 878.727 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 879 \cdot 254/2$$

$$= 111598.336 \text{ mm}^2$$



Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 717477 / (2668 \cdot 254)$$

$$= 1.059 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 111598 \cdot 1.06$$

$$= 118153.375 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 2756 + 7030$$

$$= 9785.816 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 118153 / 9786$$

$$= 12.075 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 228 - 16$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 9786$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228 / 2$$

$$= 114.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(12840037 / 9786)}$$

$$= 36.223 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(120/2) (1500 + 12 + 10) - 32$$

$$= 967.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202713 / 207)}$$

$$= 139.035$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 967 / 36.2$$

$$= 26.696$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 112286 / (2 \cdot 2668) \cdot 879 \cdot 967 / 2$$

$$= 8944102.000 \text{ N-mm}$$



Compressive Allowable, $KL/r < C_c$ ($26.7 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (26.7)^2 / (2 \cdot 139^2)) 207 / (5/3 + 3 \cdot (26.7) / (8 \cdot 139) - (26.7)^3 / (8 \cdot 139^3))$$

$$= 116.9 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 12.1/117 + (8944102 \cdot 114 / 12840049) / 138$$

$$= 0.679$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/16816$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1760 - \sqrt{ (1522 + 10)^2 - (879/2)^2 } - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 13135538 / 16816 }$$

$$= 27.949 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 271/27.9$$

$$= 9.689$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 112286 / (2 \cdot 2668)$$

$$= 21.043 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 21 \cdot 879 \cdot 271$$

$$= 5009361.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5009361 / 115224$$

$$= 43.461 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ ($9.69 < 139$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 139^2)) 207 / (5/3 + 3 \cdot (9.69) / (8 \cdot 139) - (9.69)^3 / (8 \cdot 139^3))$$

$$= 122 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 14/122 + (5009361 \cdot 114 / 13135538) / 138$$



= 0.430

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (280.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{1506 \cdot 12}, 2 \cdot 720)$$

$$= 437.7142 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1506.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	12000.00	mm
Distance from Saddle to Vessel tangent	a1 or a	720.00	mm
Saddle Width	b1 or b	228.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	754.00	mm
Shell Allowable Stress used in Calculation		108.20	MPa
Head Allowable Stress used in Calculation		108.20	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		711586.50	N
Pressure used in Analysis	P	-0.100	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	-28.64	-93.67
Long. Stress at Top of Midspan	28.64	108.20
Long. Stress at Bottom of Midspan	13.58	108.20
Long. Stress at Top of Saddles	-6.02	-112.33
Long. Stress at Top of Saddles	6.02	108.20
Long. Stress at Bottom of Saddles	-6.53	-112.33
Long. Stress at Bottom of Saddles	6.53	108.20
Tangential Shear in Shell	34.65	86.56
Tangential Shear in Head	51.97	86.56
Circ. Stress at Horn of Saddle	132.21	135.25
Addl. Stress in Head as Stiffener	5.04	135.25
Circ. Compressive Stress in Shell	10.30	108.20

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(112286, 0) \cdot 1760 / 10510$$

$$= 18803.4 \text{ N}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3(112286 / 2 + 0) \cdot 1760 / 2636$$

$$= 112448.8 \text{ N}$$

Load Combination Results for Q + Wind or Seismic [Q]:



= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 599138 + max(0, 0, 18803, 112449)
 = 711586.5 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	717727.69	N
Transverse Shear Load	56143.07	N
Longitudinal Shear Load	112286.12	N

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(3000/2 + 12 + 10) \sin(120/2), 2 \cdot 1506)$
 = 2636.181 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 = $-711586 \cdot 720 [1 - (1 - 720/12000 + (1506^2 - 754^2)/(2 \cdot 720 \cdot 12000)) / (1 + (4 \cdot 754)/(3 \cdot 12000))]$
 = -21483228.0 N-mm

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 \cdot ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 = $711586 \cdot 12000/4 \cdot ((1 + 2(1506^2 - 754^2)/(12000^2)) / (1 + (4 \cdot 754)/(3 \cdot 12000)) - 4 \cdot 720/12000)$
 = 1504501376.0 N-mm

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 = $-0.1 \cdot 1506/(2 \cdot 10) - 1504501376/(\pi \cdot 1506^2 \cdot 10)$
 = -28.64 MPa

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 = $-0.1 \cdot 1506/(2 \cdot 10) + 1504501376/(\pi \cdot 1506^2 \cdot 10)$
 = 13.58 MPa

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

= $P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$
 = $-0.1 \cdot 1506/(2 \cdot 12) - 21483228/(\pi \cdot 1506^2 \cdot 12)$
 = -6.02 MPa

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

= $P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 t)$
 = $-0.1 \cdot 1506/(2 \cdot 12) + 21483228/(\pi \cdot 1506^2 \cdot 12)$
 = -6.53 MPa

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot h_m)/3)$
 = $711586 (12000 - 2 \cdot 720) / (12000 + (4 \cdot 754/3))$
 = 577790.1 N

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

= $K3 \cdot Q / (R_m \cdot t)$
 = $0.88 \cdot 711586 / (1506 \cdot 12)$
 = 34.65 MPa

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

= $K3 \cdot Q / (R_m \cdot t_h)$
 = $0.88 \cdot 711586 / (1506 \cdot 8)$
 = 51.97 MPa



Membrane stress in the head as a stiffener (4.15.16) [σ_5]:

$$= K_4 \cdot Q / (R_m \cdot t_h) + ((P \cdot R_i) / (2 \cdot t_h)) \cdot (R_i / h_m)$$

$$= 0.4 \cdot 711586 / (1506 \cdot 8) + ((-0.1 \cdot 1500) / (2 \cdot 8)) \cdot (1500 / 754)$$

$$= 5.04 \text{ MPa}$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506 \cdot 12)}$$

$$= 104.857 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 711586 \cdot 0.1 / (12(228 + 105 + 105))$$

$$= -10.30 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -711586 / (4 \cdot 12(228 + 105 + 105)) - 12 \cdot 0.013 \cdot 711586 \cdot 1506 / (12000 \cdot 12^2)$$

$$= -132.21 \text{ MPa}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(228 + 1.56 \cdot \sqrt{(1506 \cdot 12)}, 2 \cdot 720)$$

$$= 437.71 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2668.0002	mm
Baseplate Thickness	Bpthk	32.0000	mm
Baseplate Width	Bpwid	254.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	13.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	207.0	MPa
Height of Web at Center	Hw,c	216.0	mm
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500 \cdot 12)} + 280)^3) / 12$$

$$= 70458.555 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 70459 + 5872(123 - 6)^2$$

$$= 80877888.000 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm ²	AY mm ³	I + AD ² (2) mm ⁴
Shell	489.3	12.0	6.0	5871.6	35229.3	0.809E+08
Wearplate	280.0	10.0	17.0	2800.0	47600.0	0.317E+08
Web	16.0	206.0	125.0	3296.0	411999.6	0.117E+08
BasePlate	254.0	32.0	244.0	8128.0	1983230.9	0.119E+09
Totals	...	...	...	20095.5	2478059.8	0.243E+09

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2478059 / 20096$$

$$= 123.314 \text{ mm}$$

Angle [β]:



$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.2 \cdot 711586 \\
 &= 144823.2812 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 10.1825 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 124.1978 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1760 - 1500 \cdot \sin(120/2) / 1.05 \\
 &= 519.510 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 75267640.0000 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 38.1375 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 137.9975 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(711586 + 6141)254 / (4 \cdot 2668 \cdot 138)} \\
 &= 19.272 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (1500 + 12 + 10) \\
 &= 2636.181 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 2636 / (4 - 1) \\
 &= 878.727 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 879 \cdot 254 / 2 \\
 &= 111598.336 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 717728 / (2668 \cdot 254) \\
 &= 1.059 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 111598 \cdot 1.06 \\
 &= 118194.664 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 2756 + 7030 \\
 &= 9785.816 \text{ mm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / Ar \\
 &= 118195 / 9786 \\
 &= 12.079 \text{ MPa}
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	13.0	228.0	...	2964.0	...	0.128E+08

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 228 - 16
 = 212.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/9786
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 228/2
 = 114.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{12840037/9786}$
 = 36.223 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1760 - cos(120/2)(1500 + 12 + 10) - 32
 = 967.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 202713/207}$
 = 139.035

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 967/36.2
 = 26.696

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 112286/(2 • 2668) • 879 • 967/2
 = 8944102.000 N-mm

Compressive Allowable, KL/r < Cc (26.7 < 139) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (26.7)^2 / (2 \cdot 139^2)) 207 / (5/3 + 3 \cdot (26.7) / (8 \cdot 139) - (26.7)^3 / (8 \cdot 139^3))$
 = 116.9 MPa

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 12.1/117 + (8944102 • 114/12840049) / 138
 = 0.679

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	13.0	212.0	0.0	2756.0	0.0	0.128E+08
Web	878.7	16.0	0.0	14059.6	0.0	0.300E+06
Totals	...	...	...	16815.6	...	0.131E+08

Distance to Centroid from Datum [ytot]:



$$= AY / A$$

$$= 0/16816$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 228/2$$

$$= 114.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1760 - \sqrt{(1522 + 10)^2 - (879/2)^2} - 32$$

$$= 270.796 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(13135538/16816)}$$

$$= 27.949 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 271/27.9$$

$$= 9.689$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 112286/(2 \cdot 2668)$$

$$= 21.043 \text{ N/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 21 \cdot 879 \cdot 271$$

$$= 5009361.000 \text{ N-mm}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5009361/115224$$

$$= 43.461 \text{ MPa}$$

Compressive Allowable, $KL/r < C_c$ (9.69 < 139) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.69)^2 / (2 \cdot 139^2)) 207 /$$

$$(5/3 + 3 \cdot (9.69) / (8 \cdot 139) - (9.69)^3 / (8 \cdot 139^3))$$

$$= 122 \text{ MPa}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 14.1/122 + (5009361 \cdot 114/13135538) / 138$$

$$= 0.430$$

*Review notes about nozzle loadings and supports in the Warnings report.***PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.**



Input, Nozzle Desc: N1-6"

From: 10

Pressure for Reinforcement Calculations	P	0.5485	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3000.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	c	0.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	750.0000	mm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)	Outside		
Layout Angle	90.00	deg	
Diameter	6.0000	in.	

Size and Thickness Basis	Minimum		
Nominal Thickness	80		

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

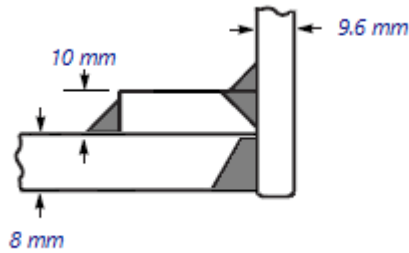
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	720.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	108.20	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	6.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		65.8625	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N1-6''

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 0.55 \cdot 3000 \cdot 0.9 / (2 \cdot 108 \cdot 1 - 0.2 \cdot 0.55)$
 $= 6.8474 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 84.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.3418 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0050 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.4494 mm
Parallel to Vessel Wall, opening length	d	155.2247 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, Sn/Sv)$
 $= \min(1, 135/108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, Sn/Sv)$
 $= \min(1, 135/108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, Sp/Sv)$
 $= \min(1, 108/108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

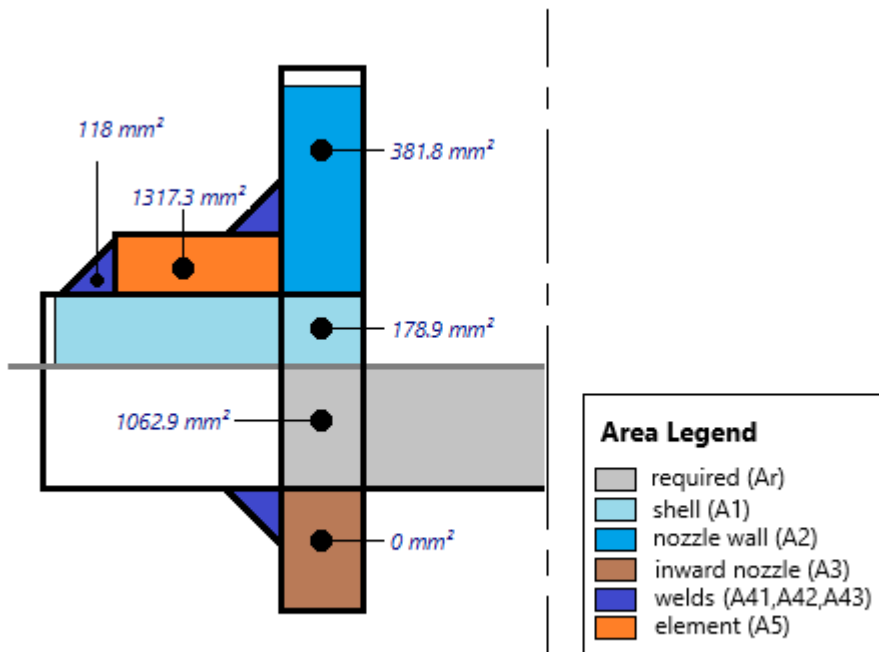


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1062.885	618.045	NA
Area in Shell	A1	178.913	5.708	NA
Area in Nozzle Wall	A2	381.821	354.470	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	117.958	117.958	NA
Area in Element	A5	1317.250	1317.250	NA
TOTAL AREA AVAILABLE	Atot	1995.942	1795.386	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 73.81 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	206.6943	10.0000 mm
Based on given Pad Diameter:	300.0000	2.9166 mm
Based on Shell or Nozzle Thickness:	223.2437	8.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (155 \cdot 6.85 \cdot 1 + 2 \cdot 9.6 \cdot 6.85 \cdot 1 \cdot (1 - 1))$$

$$= 1062.885 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 155 (1 \cdot 8 - 1 \cdot 6.85) - 2 \cdot 9.6$$

$$(1 \cdot 8 - 1 \cdot 6.85) \cdot (1 - 1)$$

$$= 178.913 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$



$$= (2 \cdot 20) \cdot (9.6 - 0.34) \cdot 1/\sin(75.9)$$

$$= 381.821 \text{ mm}^2$$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot Fr_3 + ((W_i - \text{can}/0.707)^2 - A_r \text{ Lost}) \cdot fr_2 + \text{Trap} fr_4$$

$$= (100) \cdot 1 + (0) \cdot 1 + 18^2 \cdot 1$$

$$= 117.958 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (307 - 175) \cdot 10 \cdot 1$$

$$= 1317.250 \text{ mm}^2$$

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N1-6"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6012	mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	5.0000 = $0.5 \cdot T_{minPad}$	4.2420 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (1063 - 179 + 2 \cdot 9.6 \cdot 1 \cdot (1 \cdot 8 - 6.85)) \cdot 108)$$

$$= 98032.80 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - \text{Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (382 + 1317 + 118 - 0 \cdot 1) \cdot 108$$

$$= 196587.08 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (382 + 0 + 100 + (154)) \cdot 108$$

$$= 68749.20 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (382 + 0 + 118 + 1317 + (154)) \cdot 108$$

$$= 213207.38 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 175 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 145912. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 300 \cdot 6 \cdot 0.49 \cdot 108$$

$$= 149893. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 82.6) \cdot (9.6 - 0) \cdot 0.7 \cdot 135$$



= 235146. N

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot \text{Seg}$$

$$= (3.14/2) \cdot 175 \cdot 10 \cdot 0.74 \cdot 135$$

$$= 274574. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 175 \cdot (8 - 0) \cdot 0.74 \cdot 135$$

$$= 219660. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (149893 + 235146) = 385039 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (145912 + 274574 + 219660 + 0) = 640146 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (149893 + 219660 + 0) = 369553 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 385039 N	, must exceed W = 98032 N	or W1 = 196587 N
Path 2-2 = 640146 N	, must exceed W = 98032 N	or W2 = 68749 N
Path 3-3 = 369552 N	, must exceed W = 98032 N	or W3 = 213207 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.55 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N1-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 0.55 \cdot 3000 \cdot 0.9 / (2 \cdot 108 \cdot 1 - 0.2 \cdot 0.55)$$

$$= 6.8474 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 0.55 \cdot 84.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$$

$$= 0.3418 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 1.0050 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.1452 mm
Parallel to Vessel Wall, opening length	d	149.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		20.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

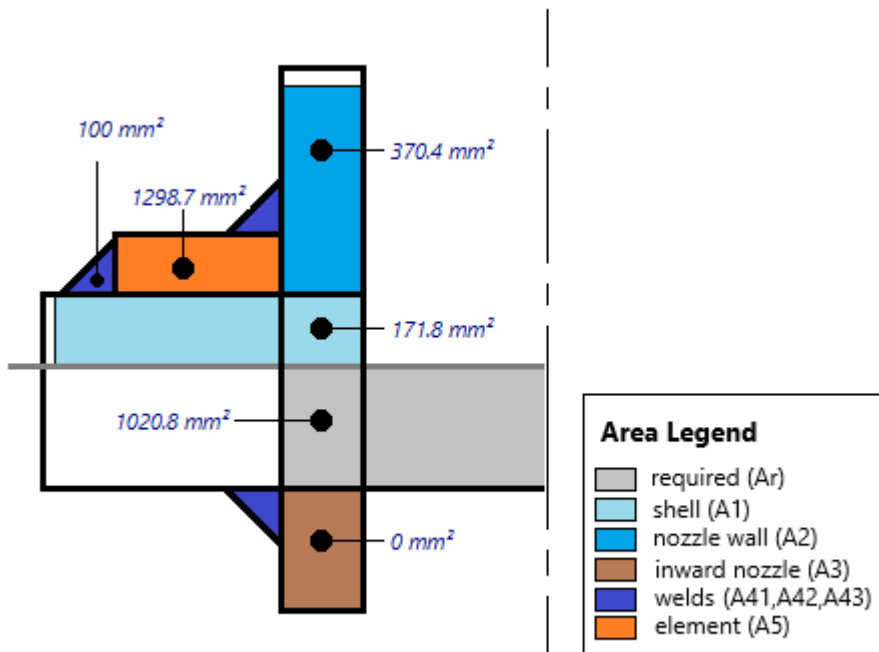


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1020.759	593.549	NA
Area in Shell	A1	171.822	5.482	NA
Area in Nozzle Wall	A2	370.378	343.847	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	1298.702	1298.702	NA
TOTAL AREA AVAILABLE	Atot	1940.901	1748.030	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	206.1309	10.0000 mm
Based on given Pad Diameter:	300.0000	2.9149 mm
Based on Shell or Nozzle Thickness:	215.5949	8.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (149 \cdot 6.85 \cdot 1 + 2 \cdot 9.6 \cdot 6.85 \cdot 1 \cdot (1 - 1))$
 $= 1020.759 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 149 (1 \cdot 8 - 1 \cdot 6.85) - 2 \cdot 9.6$
 $(1 \cdot 8 - 1 \cdot 6.85) \cdot (1 - 1)$
 $= 171.822 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 20) \cdot (9.6 - 0.34) \cdot 1$$

$$= 370.378 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot fr_4$$

$$= (298 - 168) \cdot 10 \cdot 1$$

$$= 1298.702 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.0050 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 7.6082 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 7.6082 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 1.3864 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.22, \max(7.61, 1.5)]$$

$$= 6.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.01, 6.22)$$

$$= 6.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.3,	Allowable	:	134.8 MPa	Passed
Expansion	:	0.0,	Allowable	:	314.3 MPa	Passed
Occasional	:	2.0,	Allowable	:	179.3 MPa	Passed
Shear	:	18.2,	Allowable	:	94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N1-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.6012	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$4.2420 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (1021 - 172 + 2 \cdot 9.6 \cdot 1 \cdot (1 \cdot 8 - 6.85)) \cdot 108)$$

$$= 94242.31 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - Can/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (370 + 1299 + 100 - 0 \cdot 1) \cdot 108$$

$$= 191399.34 \text{ N}$$



Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (370 + 0 + 100 + (154)) \cdot 108$$

$$= 67511.13 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (370 + 0 + 100 + 1299 + (154)) \cdot 108$$

$$= 208019.66 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 140129. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 300 \cdot 6 \cdot 0.49 \cdot 108$$

$$= 149893. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 79.3) \cdot (9.6 - 0) \cdot 0.7 \cdot 135$$

$$= 225826. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 168 \cdot 10 \cdot 0.74 \cdot 135$$

$$= 263692. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 168 \cdot (8 - 0) \cdot 0.74 \cdot 135$$

$$= 210954. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (149893 + 225826) = 375719 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (140129 + 263692 + 210954 + 0) = 614775 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (149893 + 210954 + 0) = 360847 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 375719 N	, must exceed W = 94242 N	or W1 = 191399 N
Path 2-2 = 614774 N	, must exceed W = 94242 N	or W2 = 67511 N
Path 3-3 = 360846 N	, must exceed W = 94242 N	or W3 = 208019 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.55 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 25.7554 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 754.0804 mm

Warning:



The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N1-6'' :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	0.0000	mm
Vessel Diameter	Dv	5400.000	mm
Vessel Thickness	Tv	8.000	mm
Design Temperature	T1	120.0	C
Vessel Material		SA-240 304	
Vessel UNS Number		S30400	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	108.20	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	168.275	mm
Nozzle Thickness	Tn	9.601	mm
Nozzle Material		SA-312 TP304	
Nozzle UNS Number		S30400	
Nozzle Cold S.I. Allowable	SNmc	137.64	MPa
Nozzle Hot S.I. Allowable	SNmh	134.82	MPa
Thickness of Reinforcing Pad	Tpad	10.000	mm
Diameter of Reinforcing Pad	Dpad	300.000	mm
Design Internal Pressure	Dp	0.549	MPa
Include Pressure Thrust		No	

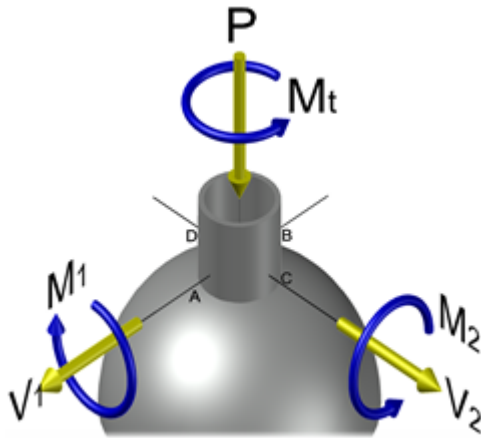
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9599.6	N
Longitudinal Shear (SUS)	(V1) V1	8484.9	N
Circumferential Shear (SUS)	(Vc) V2	8484.9	N
Circumferential Moment (SUS)	(Mc) M1	3340974.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3340974.0	N-mm
Torsional Moment (SUS)	Mt	4321524.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(2704.0 (8.0 - 0))}$$

$$= 653.633 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-9599.6	N
Circumferential Shear	(VC)	V2	8484.9	N
Longitudinal Shear	(VL)	V1	8484.9	N
Circumferential Moment	(MC)	M1	3340974.0	N-mm
Longitudinal Moment	(ML)	M2	3340974.0	N-mm
Torsional Moment		MT	4321524.5	N-mm

Dimensionless Parameters: U = 0.38 TAU = 8.26 RHO = 1.87

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.08378	(A,B,C,D)
$M(x) / P$	SP 2	0.07483	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.20483	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.21244	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.20483	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.21244	(A,B,C,D)
$N(y) * T / P$	SP 2	0.21947	(A,B,C,D)
$M(y) / P$	SP 2	0.06273	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.12669	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.24074	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.12669	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.24074	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	2.5	2.5	2.5	2.5	2.5	2.5	2.5	2.5
Rad.	Bend. P	13.3	-13.3	13.3	-13.3	13.3	-13.3	13.3	-13.3
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-9.6	-9.6	9.6	9.6
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-59.5	59.5	59.5	-59.5
Rad.	Memb. ML	-9.6	-9.6	9.6	9.6	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-59.5	59.5	59.5	-59.5	0.0	0.0	0.0	0.0

Tot. Rad. Str.	-53.3	39.1	84.9	-60.8	-53.3	39.1	84.9	-60.8
Tang. Memb. P	6.5	6.5	6.5	6.5	6.5	6.5	6.5	6.5
Tang. Bend. P	11.2	-11.2	11.2	-11.2	11.2	-11.2	11.2	-11.2
Tang. Memb. MC	0.0	0.0	0.0	0.0	-5.9	-5.9	5.9	5.9
Tang. Bend. MC	0.0	0.0	0.0	0.0	-67.4	67.4	67.4	-67.4
Tang. Memb. ML	-5.9	-5.9	5.9	5.9	0.0	0.0	0.0	0.0
Tang. Bend. ML	-67.4	67.4	67.4	-67.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-55.7	56.9	91.0	-66.2	-55.7	56.9	91.0	-66.2
Shear VC	1.8	1.8	-1.8	-1.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Tot. Shear	7.2	7.2	3.6	3.6	3.6	3.6	7.2	7.2
Str. Int.	61.8	59.4	92.7	68.0	58.3	57.6	95.7	71.1

Unitless Prm: U = 1.02 TAU = 0.00 (15.12) RHO = 0.00 (0.83)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.06412	(A,B,C,D)
M(x) / P	SR 2	0.03356	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.07439	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.06901	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.07439	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.06901	(A,B,C,D)
N(y) * T / P	SR 2	0.01942	(A,B,C,D)
M(y) / P	SR 2	0.01036	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.02205	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.02082	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.02205	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.02082	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	9.6	9.6	9.6	9.6	9.6	9.6	9.6	9.6
Rad.	Bend. P	30.2	-30.2	30.2	-30.2	30.2	-30.2	30.2	-30.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-26.4	-26.4	26.4	26.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-146.9	146.9	146.9	-146.9
Rad.	Memb. ML	-26.4	-26.4	26.4	26.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-146.9	146.9	146.9	-146.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-133.5	99.9	213.1	-141.1	-133.5	99.9	213.1	-141.1
Tang. Memb. P		2.9	2.9	2.9	2.9	2.9	2.9	2.9	2.9
Tang.	Bend. P	9.3	-9.3	9.3	-9.3	9.3	-9.3	9.3	-9.3
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-7.8	-7.8	7.8	7.8
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-44.3	44.3	44.3	-44.3
Tang.	Memb. ML	-7.8	-7.8	7.8	7.8	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-44.3	44.3	44.3	-44.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-39.9	30.1	64.4	-42.9	-39.9	30.1	64.4	-42.9
Shear VC		2.3	2.3	-2.3	-2.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Shear MT		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8



Tot. Shear	6.1	6.1	1.6	1.6	1.6	1.6	6.1	6.1
Str. Int.	133.9	100.5	213.2	141.1	133.5	100.0	213.4	141.5

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (MPa)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		41.1	41.1	41.1	41.1	41.1	41.1	41.1	41.1
Rad. Pl (SUS)		-7.1	-7.1	12.0	12.0	-7.1	-7.1	12.0	12.0
Rad. Q (SUS)		-46.2	46.2	72.8	-72.8	-46.2	46.2	72.8	-72.8
Long. Pm (SUS)		41.1	41.1	41.1	41.1	41.1	41.1	41.1	41.1
Long. Pl (SUS)		0.6	0.6	12.4	12.4	0.6	0.6	12.4	12.4
Long. Q (SUS)		-56.3	56.3	78.6	-78.6	-56.3	56.3	78.6	-78.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear Q (SUS)		5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
Pm (SUS)		41.1	41.1	41.1	41.1	41.1	41.1	41.1	41.1
Pm+Pl (SUS)		42.1	42.1	55.2	55.2	42.1	42.1	55.2	55.2
Pm+Pl+Q (Total)		20.6	100.5	133.8	26.8	17.1	98.7	136.9	30.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.14	108.20	Passed
Pm+Pl (SUS)	55.17	162.30	Passed
Pm+Pl+Q (TOTAL)	136.87	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (MPa)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		92.6	92.6	92.6	92.6	92.6	92.6	92.6	92.6
Rad. Pl (SUS)		-16.8	-16.8	36.0	36.0	-16.8	-16.8	36.0	36.0
Rad. Q (SUS)		-116.7	116.7	177.1	-177.1	-116.7	116.7	177.1	-177.1
Long. Pm (SUS)		92.6	92.6	92.6	92.6	92.6	92.6	92.6	92.6
Long. Pl (SUS)		-4.9	-4.9	10.7	10.7	-4.9	-4.9	10.7	10.7
Long. Q (SUS)		-35.0	35.0	53.6	-53.6	-35.0	35.0	53.6	-53.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.3	2.3	-2.3	-2.3	-2.3	-2.3	2.3	2.3
Shear Q (SUS)		3.8	3.8	3.8	3.8	3.8	3.8	3.8	3.8
Pm (SUS)		92.6	92.6	92.6	92.6	92.6	92.6	92.6	92.6
Pm+Pl (SUS)		88.1	88.1	128.8	128.8	88.1	88.1	128.8	128.8
Pm+Pl+Q (Total)		94.4	193.0	305.7	98.3	93.6	192.5	306.0	99.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.56	108.20	Passed
Pm+Pl (SUS)	128.77	162.30	Passed
Pm+Pl+Q (TOTAL)	305.95	324.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: N9-6"****From: 20**

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2649.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		300.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

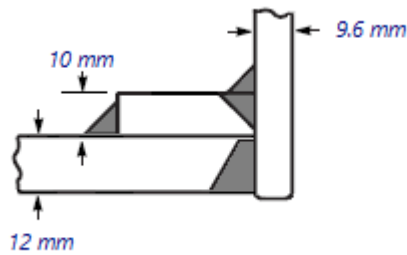
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	108.20	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		65.8625	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N9-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 84.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.3406 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3893 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.1452 mm
Parallel to Vessel Wall, opening length	d	149.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

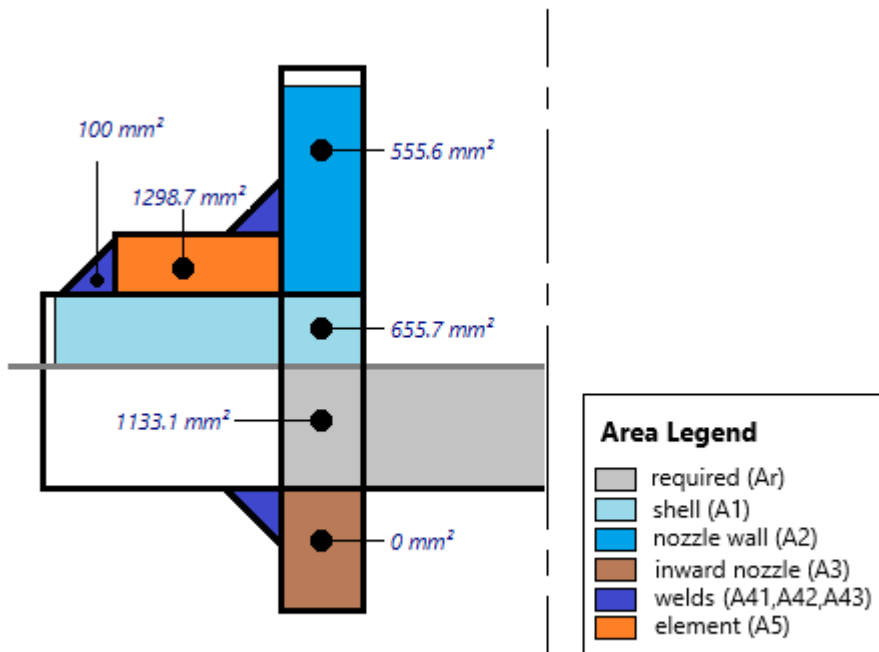


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1133.147	693.564	NA
Area in Shell	A1	655.724	401.743	NA
Area in Nozzle Wall	A2	555.637	492.715	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	1298.702	1298.702	NA
TOTAL AREA AVAILABLE	Atot	2610.063	2293.161	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (149 \cdot 7.6 \cdot 1 + 2 \cdot 9.6 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 1133.147 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 149 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 9.6$$

$$(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 655.724 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 30) \cdot (9.6 - 0.34) \cdot 1$$

$$= 555.637 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot f_{r4}$$

$$= (298 - 168) \cdot 10 \cdot 1$$

$$= 1298.702 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.3893 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 7.6013 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 7.6013 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 1.3871 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 6.2200 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[6.22, \max(7.6, 1.5)]$$

$$= 6.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1.39, 6.22)$$

$$= 6.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N9-6"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.6012 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (1133 - 656 + 2 \cdot 9.6 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 60791.49 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - Can / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (556 + 1299 + 100 - 0 \cdot 1) \cdot 108$$

$$= 211442.80 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (556 + 0 + 100 + (230)) \cdot 108$$

$$= 95864.74 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (556 + 0 + 100 + 1299 + (230)) \cdot 108$$

$$= 236373.23 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 140129. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 300 \cdot 8 \cdot 0.49 \cdot 108$$

$$= 199857. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 79.3) \cdot (9.6 - 0) \cdot 0.7 \cdot 135$$

$$= 225826. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 10 \cdot 0.74 \cdot 135$$

$$= 263692. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 263692. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (199857 + 225826) = 425684 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (140129 + 263692 + 263692 + 0) = 667513 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (199857 + 263692 + 0) = 463550 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 425683 N	, must exceed W = 60791 N	or W1 = 211442 N
Path 2-2 = 667513 N	, must exceed W = 60791 N	or W2 = 95864 N
Path 3-3 = 463549 N	, must exceed W = 60791 N	or W3 = 236373 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.83 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 2.3616 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1504.3617 mm

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Input, Nozzle Desc: N2-3"

From: 20

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2649.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		700.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

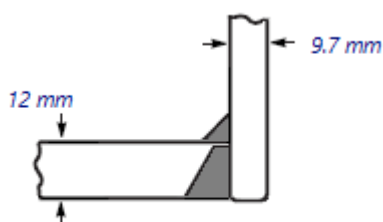
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N2-3"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 3.500 in.
 Actual Thickness Used in Calculation 0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 44.5 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.1799 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7894 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.8618	mm
Parallel to Vessel Wall, opening length	d	69.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.3364	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N2-3".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.7894	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.6013	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.6013	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.3871	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(7.6, 1.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(0.79, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.1,	Allowable	:	134.8 MPa	Passed
Expansion	:	0.0,	Allowable	:	314.4 MPa	Passed
Occasional	:	0.9,	Allowable	:	179.3 MPa	Passed
Shear	:	20.3,	Allowable	:	94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C



Weld Size Calculations, Description: N2-3"

Intermediate Calc. for nozzle/shell Welds Tmin 9.7345 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.72 MPa

The Drop for this Nozzle is : 0.6588 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1502.6588 mm

Input Echo, WRC297 Item 1, Description: N2-3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	88.900	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 137.64	MPa
Nozzle Hot S.I. Allowable	SNmh 134.82	MPa

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.132) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1506.0(12.0 - 0.0))}$
 $= 532.526 \text{ mm}$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 3012.000 mm
 Nozzle Outside Diameter (d) = 88.900 mm
 Vessel Thickness used (T) = 12.000 mm
 Nozzle Thickness used (t) = 9.735 mm
 T / t = 1.233
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.468$$

Nr/P = 0.187
 Mr/P = 0.268
 M0/P = 0.105
 N0/P = 0.174
 MrD/Mc = 0.276
 NrDL/Mc = 0.156
 M0d/Mc = 0.131
 N0DL/Mc = -0.105
 MrD/M1 = 0.253
 NrDL/M1 = 0.152
 M0D/M1 = 0.117
 N0DL/M1 = -0.024

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-51	51	51	-51
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	45	106	144	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-110	110	110	-110
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	-28	86	215	-112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7
Shear Stress Summary	9	9	5	5

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	75	110	216	113



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	5	5	-5	-5
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	22	68	99	-31

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-93	93	93	-93
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	25	105	229	-62

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	7	7	7	7
Shear Stress Summary	4	4	10	10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	106	230	65

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	9	10	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-117	117	117	-117
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-83	46	195	-146
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8

Shear Stress Summary	11	11	6	6
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	96	49	195	153

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	5	5	-5	-5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2

Hoop Stress Summary	13	14	1	2
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-101	101	101	-101
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-62	34	173	-134
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	8	8	8	8

Shear Stress Summary	5	5	12	12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress 76 36 174 139

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	6	6	4	4	-2	-2	14	14
Circ. Q (SUS)	-30	30	72	-72	-40	40	146	-146
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-5	-5	17	17	10	10	0	0
Long. Q (SUS)	-57	57	163	-163	-23	23	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	74.1	74.1	72.2	72.2	66.2	66.2	82.1	82.1
Pm+Pl+Q (Total)	74.2	107.7	214.4	112.4	29.1	106.6	228.6	66.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	108.20	Passed
Pm+Pl (SUS)	82.08	162.30	Passed
Pm+Pl+Q (TOTAL)	228.62	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	6	6	4	4	10	10	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	22	22	-15	-15	17	17
Long. Q (SUS)	-65	65	169	-169	-49	49	153	-153
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	27.3	27.3	23.2	23.2	26.7	26.7	18.5	18.5
Pm+Pl+Q (Total)	94.1	48.5	192.2	152.5	75.7	36.0	171.7	138.8

Nozzle Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	134.82	Passed
Pm+Pl (SUS)	27.29	202.23	Passed
Pm+Pl+Q (TOTAL)	192.19	404.47	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N3-6"
From: 20

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2649.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

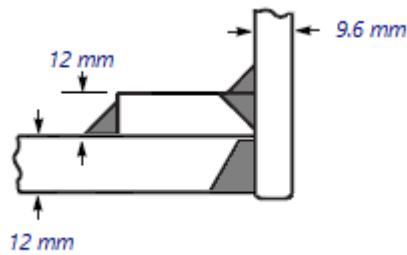
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304L	
Pad Allowable Stress at Temperature	Sp	115.00	MPa
Pad Allowable Stress At Ambient	Spa	115.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		65.8625	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N3-6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 84.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.3406 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3893 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.1452 mm
Parallel to Vessel Wall, opening length	d	149.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

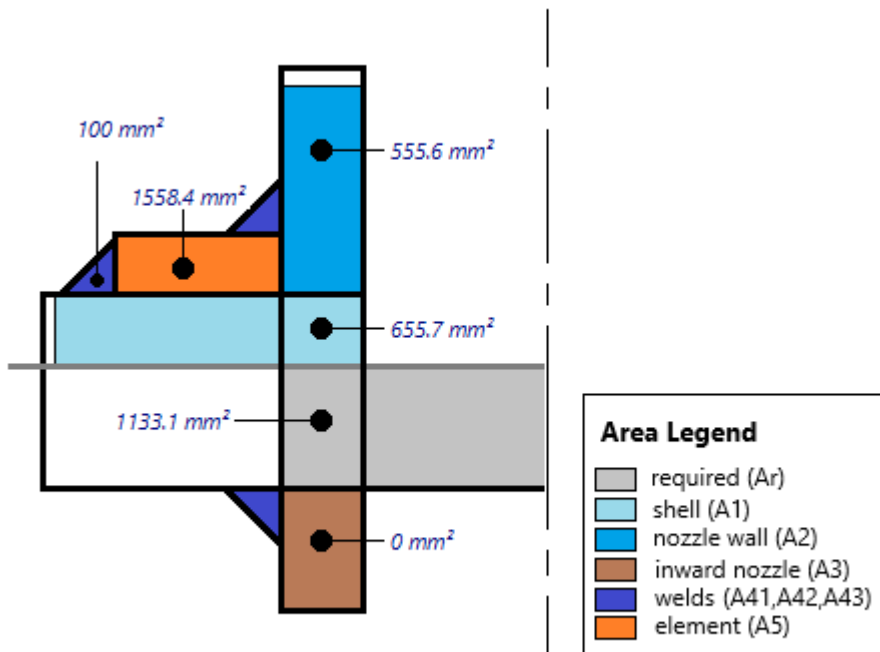


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1133.147	693.564	NA
Area in Shell	A1	655.724	401.743	NA
Area in Nozzle Wall	A2	555.637	492.715	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	1558.442	1558.442	NA
TOTAL AREA AVAILABLE	Atot	2869.803	2552.901	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (149 \cdot 7.6 \cdot 1 + 2 \cdot 9.6 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 1133.147 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 149 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 9.6$$

$$(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 655.724 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 30) \cdot (9.6 - 0.34) \cdot 1$$

$$= 555.637 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot f_r4$$

$$= (298 - 168) \cdot 12 \cdot 1$$

$$= 1558.442 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.3893 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.6013 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.6013 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3871 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 6.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.22, \max(7.6, 1.5)]$$

$$= 6.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.39, 6.22)$$

$$= 6.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.3,	Allowable	: 134.8 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.3 MPa	Passed
Occasional	: 2.0,	Allowable	: 179.3 MPa	Passed
Shear	: 18.2,	Allowable	: 94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N3-6"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6012 mm
Intermediate Calc. for pad/shell Welds	TminPad	12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5 * TminPad	5.6560 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1133 - 656 + 2 \cdot 9.6 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 60791.49 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (556 + 1558 + 100 - 0 \cdot 1) \cdot 108$$



$$= 239544.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (556 + 0 + 100 + (230)) \cdot 108$$

$$= 95864.74 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (556 + 0 + 100 + 1558 + (230)) \cdot 108$$

$$= 264474.94 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 148932. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 300 \cdot 8 \cdot 0.49 \cdot 108$$

$$= 199857. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 79.3) \cdot (9.6 - 0) \cdot 0.7 \cdot 135$$

$$= 225826. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 12 \cdot 0.74 \cdot 135$$

$$= 316431. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 263692. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (199857 + 225826) = 425684 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (148932 + 316431 + 263692 + 0) = 729055 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (199857 + 263692 + 0) = 463550 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 425683 N	, must exceed W = 60791 N	or W1 = 239544 N
Path 2-2 = 729054 N	, must exceed W = 60791 N	or W2 = 95864 N
Path 3-3 = 463549 N	, must exceed W = 60791 N	or W3 = 264474 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.83 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 2.3616 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1504.3617 mm

**Input Echo, WRC297 Item 1, Description: N3-6" :**

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	9.6012	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 137.64	MPa
Nozzle Hot S.I. Allowable	SNmh 134.82	MPa
Diameter of Reinforcing Pad	300.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-9599.58	N
Circumferential Shear	Vc	7199.67	N
Longitudinal Shear	Vl	9599.58	N
Circumferential Moment	Mc	0.288E+07	N-mm
Longitudinal Moment	Ml	0.375E+07	N-mm
Torsional Moment	Mt	0.432E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1512.0(12.0 - 0.0))}$$

$$= 612.784 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3024.000	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	24.000	mm
Nozzle Thickness used	(t) =	9.601	mm
T / t	=	2.500	
d / t	=	17.526	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.625$$

Nr/P	=	0.141
Mr/P	=	0.156
M0/P	=	0.121
N0/P	=	0.216
MrD/Mc	=	0.206
NrDL/Mc	=	0.161
M0d/Mc	=	0.188



$N\theta DL/Mc = 0.124$
 $MrD/Ml = 0.186$
 $NrDL/Ml = 0.149$
 $M\theta D/Ml = 0.166$
 $N\theta DL/Ml = 0.119$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-38	38	38	-38
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	6	59	92	-7

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-13	41	83	-33

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	23	61	93	33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-4	38	69	-21

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34
Inplane Stress Summary	10	53	93	-10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	15	53	94	23

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	3	4	12	13

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-160	160	160	-160
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-124	91	239	-188

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	10	10	10	10
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Shear Stress Summary	12	12	7	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	130	93	239	203
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5

Hoop Stress Summary	4	5	12	12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-139	139	139	-139
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-98	74	213	-172
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	103	75	214	187
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1506.0(12.0 - 0.0)}$$

$$= 611.901 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3012.000 mm

Nozzle Outside Diameter (d) = 300.000 mm

Vessel Thickness used (T) = 12.000 mm



Nozzle Thickness used (t) = 75.464 mm
 T / t = 0.159
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.578$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.090
 Mr/P = 0.172
 M0/P = 0.050
 N0/P = 0.079
 MrD/Mc = 0.299
 NrDL/Mc = 0.093
 M0d/Mc = 0.092
 N0DL/Mc = 0.062
 MrD/M1 = 0.136
 NrDL/M1 = 0.107
 M0D/M1 = 0.042
 N0DL/M1 = 0.097

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	63	67	123	40

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	68	-68	68	-68
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-70	70	70	-70
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	28	32	188	-90

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	63	67	188	131

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-36	36	36	-36
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	18	51	100	-13

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	68	-68	68	-68
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-119	119	119	-119
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	17	118	268	-107

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	18	118	268	107

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	-1	-1	7	7	-2	-2	6	6
Circ. Q (SUS)	-26	26	50	-50	-21	21	51	-51
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	7	7	0	0	6	6
Long. Q (SUS)	-28	28	58	-58	-21	21	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0



Pm+Pl (SUS)	32.1	33.0	40.1	41.1	31.1	32.1	39.1	40.1
Pm+Pl+Q (Total)	23.3	60.3	90.9	35.3	16.2	53.5	91.0	24.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	108.20	Passed
Pm+Pl (SUS)	41.06	162.30	Passed
Pm+Pl+Q (TOTAL)	91.04	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	-3	-3	13	13	-1	-1	11	11
Circ. Q (SUS)	-1	1	41	-41	-51	51	187	-187
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-4	-4	14	14	1	1	9	9
Long. Q (SUS)	-2	2	138	-138	-16	16	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	65.0	65.0	81.0	81.0	67.0	67.0	79.0	79.0
Pm+Pl+Q (Total)	64.2	66.3	186.0	130.0	19.3	118.0	266.1	108.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	108.20	Passed
Pm+Pl (SUS)	81.03	162.30	Passed
Pm+Pl+Q (TOTAL)	266.05	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-1	-1	7	7	0	0	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-19	-19	21	21	-15	-15	17	17



Long. Q (SUS)	-107	107	213	-213	-86	86	192	-192
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	20.4	21.4	23.3	23.4	18.0	19.0	19.9	20.0
Pm+Pl+Q (Total)	129.2	91.6	236.3	202.6	103.9	73.7	211.8	185.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	134.82	Passed
Pm+Pl (SUS)	23.35	202.23	Passed
Pm+Pl+Q (TOTAL)	236.28	404.47	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: M1-24"****From: 20**

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2649.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1800.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		0.00	°C
---	--	------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-240 304	
Material UNS Number		S30400	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	108.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

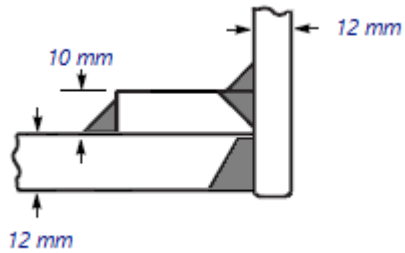
Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	108.20	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	900.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		145.2000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		150
Flange Grade		GR 2.1

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 305 / (108 \cdot 1 + 0.4 \cdot 0.55)$
 $= 1.5368 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.8815 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1171.2000 mm
Parallel to Vessel Wall, opening length	d	585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

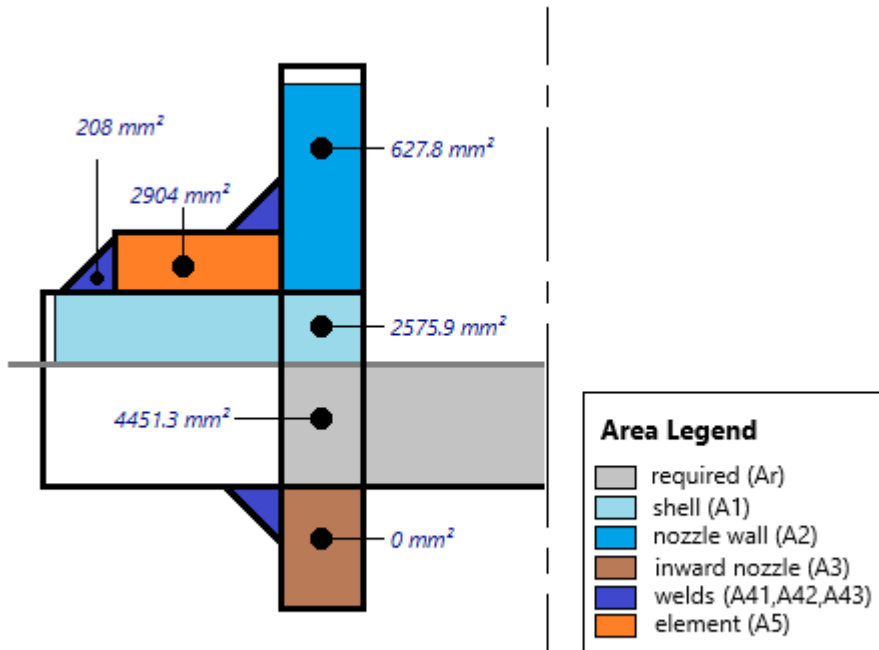


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4451.326	2724.518	NA
Area in Shell	A1	2575.874	1578.164	NA
Area in Nozzle Wall	A2	627.792	547.109	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	208.000	208.000	NA
Area in Element	A5	2904.000	2904.000	NA
TOTAL AREA AVAILABLE	Atot	6315.665	5237.273	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	713.5661	10.0000 mm
Based on given Pad Diameter:	900.0000	3.5801 mm
Based on Shell or Nozzle Thickness:	696.2384	12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (586 \cdot 7.6 \cdot 1 + 2 \cdot 12 \cdot 7.6 \cdot 1 \cdot (1 - 1))$
 $= 4451.326 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 586 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 12$
 $(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$
 $= 2575.874 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$



$$= (2 \cdot 30) \cdot (12 - 1.54) \cdot 1$$

$$= 627.792 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 12^2 \cdot 1 + (0)^2 \cdot 1 + 8^2 \cdot 1$$

$$= 208.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (900 - 610) \cdot 10 \cdot 1$$

$$= 2904.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.8815 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.8815 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: M1-24''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	8.4840 = $0.7 \cdot W_o$ mm
Pad Weld	5.0000 = $0.5 \cdot T_{minPad}$	5.6560 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (4451 - 2576 + 2 \cdot 12 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 214329.67 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (628 + 2904 + 208 - 0 \cdot 1) \cdot 108$$

$$= 404613.66 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (628 + 0 + 144 + (288)) \cdot 108$$

$$= 114660.49 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (628 + 0 + 208 + 2904 + (288)) \cdot 108$$

$$= 435772.81 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 12 \cdot 0.49 \cdot 108$$

$$= 609165. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 900 \cdot 8 \cdot 0.49 \cdot 108$$



= 599572. N

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 299) \cdot (12 - 0) \cdot 0.7 \cdot 108$$

$$= 853106. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 10 \cdot 0.74 \cdot 108$$

$$= 766637. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (10 - 0) \cdot 0.74 \cdot 108$$

$$= 766637. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (599572 + 853106) = 1452678 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (609165 + 766637 + 766637 + 0) = 2142439 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (599572 + 766637 + 0) = 1366209 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1452678 N	, must exceed W = 214329 N	or W1 = 404613 N
Path 2-2 = 2142439 N	, must exceed W = 214329 N	or W2 = 114660 N
Path 3-3 = 1366209 N	, must exceed W = 214329 N	or W3 = 435772 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.66 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 31.2941 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1533.2942 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

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Input, Nozzle Desc: N7-4''

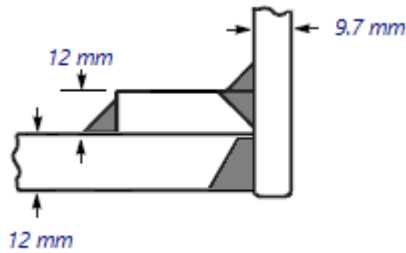
From: 20

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2449.9998	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2900.00	mm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	134.60	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-240 304L	
Pad Allowable Stress at Temperature	Sp	115.00	MPa
Pad Allowable Stress At Ambient	Spa	115.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		42.8500	mm
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N7-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 57.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.2317 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0038 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	189.6618 mm
Parallel to Vessel Wall, opening length	d	94.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

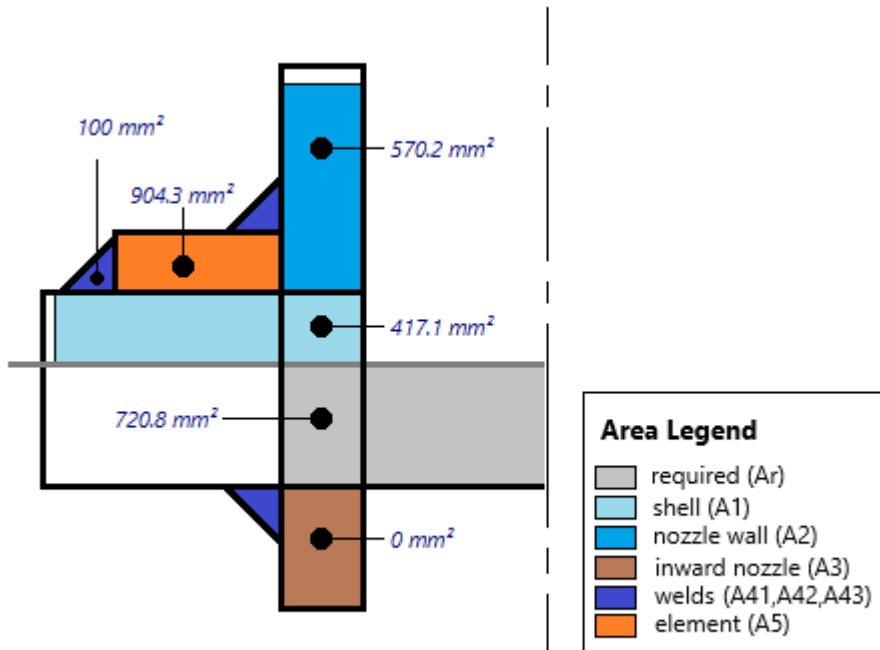


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	720.839	426.536	NA
Area in Shell	A1	417.132	284.898	NA
Area in Nozzle Wall	A2	570.169	523.847	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	904.342	904.342	NA
TOTAL AREA AVAILABLE	Atot	1991.643	1813.087	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (94.8 \cdot 7.6 \cdot 1 + 2 \cdot 9.73 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 720.839 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 94.8 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 9.73$$

$$(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 417.132 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 30) \cdot (9.73 - 0.23) \cdot 1$$

$$= 570.169 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4}$$

$$= (190 - 114) \cdot 12 \cdot 1$$

$$= 904.342 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.0038 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 7.6013 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 7.6013 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 1.3871 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 5.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[5.26, \max(7.6, 1.5)]$$

$$= 5.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1, 5.26)$$

$$= 5.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.4,	Allowable	: 134.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.3 MPa	Passed
Occasional	: 1.2,	Allowable	: 179.0 MPa	Passed
Shear	: 19.5,	Allowable	: 94.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N7-4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.7345 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$6.0000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (721 - 417 + 2 \cdot 9.73 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 42123.89 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (570 + 904 + 100 - 0 \cdot 1) \cdot 108$$



$$= 170348.69 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (570 + 0 + 100 + (234)) \cdot 108$$

$$= 97783.31 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (570 + 0 + 100 + 904 + (234)) \cdot 108$$

$$= 195625.36 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 101162. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 200 \cdot 8 \cdot 0.49 \cdot 108$$

$$= 133238. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 52.3) \cdot (9.73 - 0) \cdot 0.7 \cdot 135$$

$$= 150635. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 12 \cdot 0.74 \cdot 135$$

$$= 214577. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 178814. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (133238 + 150635) = 283874 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (101162 + 214577 + 178814 + 0) = 494553 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (133238 + 178814 + 0) = 312053 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 283873 N	, must exceed W = 42123 N	or W1 = 170348 N
Path 2-2 = 494553 N	, must exceed W = 42123 N	or W2 = 97783 N
Path 3-3 = 312052 N	, must exceed W = 42123 N	or W3 = 195625 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.83 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.0891 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1503.0891 mm

**Input Echo, WRC297 Item 1, Description: N7-4" :**

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 134.60	MPa
Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1512.0(12.0 - 0.0))}$$

$$= 558.809 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3024.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	24.000	mm
Nozzle Thickness used	(t) =	9.735	mm
T / t	=	2.465	
d / t	=	11.742	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.424$$

Nr/P	=	0.202
Mr/P	=	0.193
M0/P	=	0.148
N0/P	=	0.192
MrD/Mc	=	0.211
NrDL/Mc	=	0.186
M0d/Mc	=	0.195



$N\theta DL/Mc = -0.114$
 $MrD/Ml = 0.200$
 $NrDL/Ml = 0.182$
 $M\theta D/Ml = 0.184$
 $N\theta DL/Ml = -0.053$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	19	55	72	-2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-30	30	30	-30
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-2	31	66	-19

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	24	56	73	19

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	8	34	49	-15



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-24	24	24	-24
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34
Inplane Stress Summary	20	44	77	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	21	45	78	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	6	7	3	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-100	100	100	-100
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-81	44	162	-112

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	9	9	9	9
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Shear Stress Summary	12	12	6	6
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE

(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	90	48	162	117
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE

(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	7	7	2	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE

(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-83	83	83	-83
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-59	33	140	-100
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE

(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9

Shear Stress Summary	5	5	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE

(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	67	34	142	107
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1506.0(12.0 - 0.0)}$$

$$= 557.926 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3012.000 mm

Nozzle Outside Diameter (d) = 200.000 mm

Vessel Thickness used (T) = 12.000 mm



Nozzle Thickness used (t) = 52.585 mm
 T / t = 0.228
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.052$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.104
 Mr/P = 0.221
 M0/P = 0.064
 N0/P = 0.115
 MrD/Mc = 0.311
 NrDL/Mc = 0.072
 M0d/Mc = 0.095
 N0DL/Mc = 0.047
 MrD/M1 = 0.187
 NrDL/M1 = 0.095
 M0D/M1 = 0.059
 N0DL/M1 = 0.110

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	63	70	116	42

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	27	39	168	-79

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	64	71	168	122



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	28	45	83	-1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-82	82	82	-82
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	45	93	217	-65

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	45	93	217	65

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	3	3	1	1	-1	-1	5	5
Circ. Q (SUS)	-18	18	36	-36	-12	12	36	-36
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	6	6	4	4	0	0
Long. Q (SUS)	-18	18	42	-42	-13	13	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0



Pm+Pl (SUS)	36.0	37.0	34.1	35.1	32.1	33.1	38.0	39.0
Pm+Pl+Q (Total)	23.4	55.7	70.6	20.2	20.3	45.3	74.6	19.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	108.20	Passed
Pm+Pl (SUS)	39.04	162.30	Passed
Pm+Pl+Q (TOTAL)	74.58	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	-1	-1	11	11	1	1	7	7
Circ. Q (SUS)	-3	3	37	-37	-24	24	140	-140
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-1	-1	9	9	3	3	7	7
Long. Q (SUS)	-6	6	122	-122	-8	8	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	67.0	67.0	79.0	79.0	69.0	69.0	75.0	75.0
Pm+Pl+Q (Total)	64.2	70.3	165.0	121.0	45.1	93.0	215.1	65.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	108.20	Passed
Pm+Pl (SUS)	79.03	162.30	Passed
Pm+Pl+Q (TOTAL)	215.07	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	3	3	1	1	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	22	22	-15	-15	17	17



Long. Q (SUS)	-63	63	137	-137	-46	46	120	-120
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	24.3	25.3	23.2	23.2	20.9	21.8	18.5	18.6
Pm+Pl+Q (Total)	89.7	47.0	160.3	118.8	67.1	33.4	139.1	107.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	134.60	Passed
Pm+Pl (SUS)	25.32	201.90	Passed
Pm+Pl+Q (TOTAL)	160.31	403.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N6-4''****From: 30**

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2449.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3200.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		0.00	°C
---	--	------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

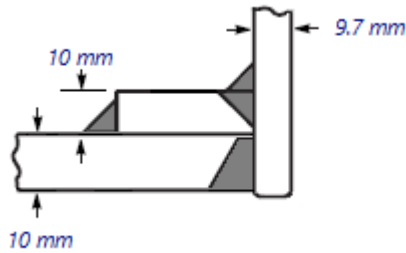
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304L	
Pad Allowable Stress at Temperature	Sp	115.00	MPa
Pad Allowable Stress At Ambient	Spa	115.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		42.8500	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N6-4"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 57.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.2313 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0038 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	189.6618 mm
Parallel to Vessel Wall, opening length	d	94.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		25.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

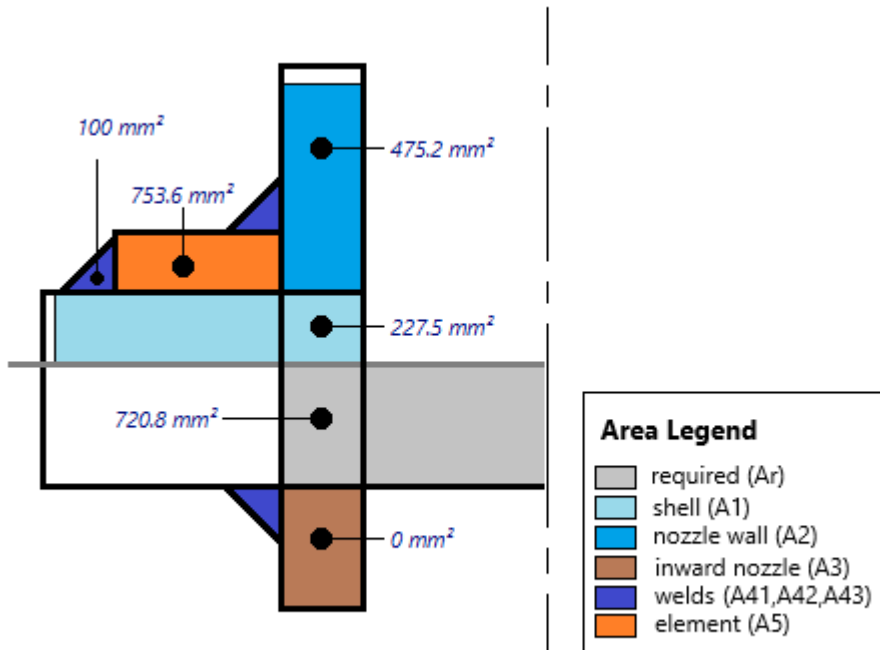


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	720.839	426.212	NA
Area in Shell	A1	227.470	95.885	NA
Area in Nozzle Wall	A2	475.160	436.539	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	753.618	753.618	NA
TOTAL AREA AVAILABLE	Atot	1556.248	1386.042	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (94.8 \cdot 7.6 \cdot 1 + 2 \cdot 9.73 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 720.839 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 94.8(1 \cdot 10 - 1 \cdot 7.6) - 2 \cdot 9.73$$

$$(1 \cdot 10 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 227.470 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 25) \cdot (9.73 - 0.23) \cdot 1$$

$$= 475.160 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (190 - 114) \cdot 10 \cdot 1$$

$$= 753.618 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 1.0038 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 7.6013 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 7.6013 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 1.3871 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 5.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[5.26, \max(7.6, 1.5)]$$

$$= 5.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(1, 5.26)$$

$$= 5.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.4,	Allowable	:	134.8 MPa	Passed
Expansion	:	0.0,	Allowable	:	314.2 MPa	Passed
Occasional	:	1.2,	Allowable	:	179.3 MPa	Passed
Shear	:	19.5,	Allowable	:	94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N6-4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.7345	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (721 - 227 + 2 \cdot 9.73 \cdot 1 \cdot (1 \cdot 10 - 7.6)) \cdot 108)$$

$$= 58430.90 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (475 + 754 + 100 - 0 \cdot 1) \cdot 108$$



$$= 143762.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (475 + 0 + 100 + (195)) \cdot 108$$

$$= 83291.35 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (475 + 0 + 100 + 754 + (195)) \cdot 108$$

$$= 164826.41 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 101162. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 200 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 166548. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 52.3) \cdot (9.73 - 0) \cdot 0.7 \cdot 135$$

$$= 150886. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 10 \cdot 0.74 \cdot 135$$

$$= 179112. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 179112. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (166548 + 150886) = 317434 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (101162 + 179112 + 179112 + 0) = 459385 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (166548 + 179112 + 0) = 345660 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 317433 N	, must exceed W = 58430 N	or W1 = 143762 N
Path 2-2 = 459384 N	, must exceed W = 58430 N	or W2 = 83291 N
Path 3-3 = 345659 N	, must exceed W = 58430 N	or W3 = 164826 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.69 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.0891 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1501.0891 mm



Input Echo, WRC297 Item 1, Description: N6-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	10.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 137.64	MPa
Nozzle Hot S.I. Allowable	SNmh 134.82	MPa
Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1510.0(10.0 - 0.0))}$$

$$= 519.811 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3020.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	20.000	mm
Nozzle Thickness used	(t) =	9.735	mm
T / t	=	2.055	
d / t	=	11.742	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.465$$

Nr/P	=	0.198
Mr/P	=	0.197
M0/P	=	0.123
N0/P	=	0.176
MrD/Mc	=	0.226
NrDL/Mc	=	0.191
M0d/Mc	=	0.176



NØDL/Mc = -0.110
 MrD/Ml = 0.211
 NrDL/Ml = 0.184
 MØD/Ml = 0.163
 NØDL/Ml = -0.044

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	21	69	89	-4

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-10	43	95	-34

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	33	70	96	35

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	3	3	-3	-3
Outplane Bending (Mc)	-29	29	29	-29
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	8	43	61	-21

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	18	-18	18	-18
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-37	37	37	-37
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	41	40	41
Inplane Stress Summary	19	58	106	-7

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	20	58	106	23

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	7	7	4	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-109	109	109	-109
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-88	51	174	-124

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	9	9	9	9
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Shear Stress Summary	12	12	6	6
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	98	55	174	129

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	3	3	-3	-3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3
Hoop Stress Summary	8	9	2	3

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-92	92	92	-92
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-65	39	152	-112

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9
Shear Stress Summary	5	5	13	13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	75	40	153	118

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1505.0(10.0 - 0.0)}$$

$$= 519.139 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3010.000 mm

Nozzle Outside Diameter (d) = 200.000 mm

Vessel Thickness used (T) = 10.000 mm



Nozzle Thickness used (t) = 52.585 mm
 T / t = 0.190
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.153$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.102
 Mr/P = 0.209
 M0/P = 0.061
 N0/P = 0.105
 MrD/Mc = 0.309
 NrDL/Mc = 0.076
 M0d/Mc = 0.094
 N0DL/Mc = 0.052
 MrD/M1 = 0.175
 NrDL/M1 = 0.100
 M0D/M1 = 0.054
 N0DL/M1 = 0.108

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	81	82	81	82
Outplane Stress Summary	75	83	147	47

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	80	-80	80	-80
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-87	87	87	-87
Inplane Pressure Stress	40	40	40	40
Inplane Stress Summary	31	46	223	-112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	76	84	223	159



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-36	36	36	-36
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	40	40	40	40
Outplane Stress Summary	31	57	110	-8

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	80	-80	80	-80
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-118	118	118	-118
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	81	82	81	82
Inplane Stress Summary	45	122	292	-105

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	45	122	292	105

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	3	3	1	1	-2	-2	8	8
Circ. Q (SUS)	-24	24	46	-46	-19	19	55	-55
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-3	-3	9	9	5	5	-1	-1
Long. Q (SUS)	-28	28	64	-64	-18	18	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0



Pm+Pl (SUS)	43.0	44.0	41.1	42.1	38.1	39.1	48.0	49.0
Pm+Pl+Q (Total)	31.6	69.0	94.2	35.3	19.7	58.6	103.6	22.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	108.20	Passed
Pm+Pl (SUS)	49.03	162.30	Passed
Pm+Pl+Q (TOTAL)	103.56	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Reinforcing Pad Edge (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	81	82	81	82	81	82	81	82
Circ. Pl (SUS)	-2	-2	14	14	2	2	10	10
Circ. Q (SUS)	-4	4	50	-50	-38	38	198	-198
Long. Pm (SUS)	40	40	40	40	40	40	40	40
Long. Pl (SUS)	-2	-2	14	14	3	3	9	9
Long. Q (SUS)	-7	7	167	-167	-13	13	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	81.0	82.0	81.0	82.0	81.0	82.0	81.0	82.0
Pm+Pl (SUS)	79.0	80.0	95.0	96.0	83.1	84.1	91.1	92.1
Pm+Pl+Q (Total)	75.4	84.4	221.1	159.1	45.1	122.0	289.1	106.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	82.00	108.20	Passed
Pm+Pl (SUS)	96.02	162.30	Passed
Pm+Pl+Q (TOTAL)	289.14	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	3	3	1	1	5	5	-1	-1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	22	22	-15	-15	17	17



Long. Q (SUS)	-70	70	148	-148	-53	53	131	-131
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	24.3	25.3	23.2	23.2	21.8	22.8	18.5	18.5
Pm+Pl+Q (Total)	96.5	53.5	171.3	129.8	75.0	40.1	150.0	117.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	134.82	Passed
Pm+Pl (SUS)	25.32	202.23	Passed
Pm+Pl+Q (TOTAL)	171.29	404.47	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1-2"

From: 30

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2449.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3500.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
---	--	------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		XS	

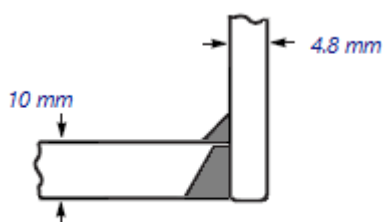
Flange Material		SA-182 F304	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K1-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.
 Actual Thickness Used in Calculation 0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 30.2 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.1221 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5399 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.2698	mm
Parallel to Vessel Wall, opening length	d	50.6349	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.1126	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.5399	mm
Wall Thickness per UG16(b),	tr16b	= 1.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.6013	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.6013	mm
Wall Thickness, shell/head, external pressure	trb2	= 1.3871	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.42, \max(7.6, 1.5)]$
 $= 3.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(0.54, 3.42)$
 $= 3.4200 \text{ mm}$

Available Nozzle Neck Thickness = 4.8450 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: K1-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	4.8450	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.3915 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

LYE DRAIN VESSEL

PV Elite 28

Licensee: SL User



FileName : A167NHM07HP001&002&0003_LYE DRAIN VESSELS -7%

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Nozzle Calcs.: K1-2''

Nozl: 36 3:43pm Sep 18,2026



HEXAGON

The Drop for this Nozzle is : 0.3033 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1500.3032 mm

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Input, Nozzle Desc: K2-4''

From: 30

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2449.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		3800.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

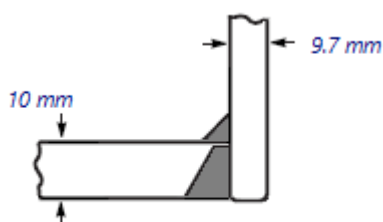
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K2-4''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1) $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$ $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1) $= 0.55 \cdot 57.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$ $= 0.2313 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0038 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 189.6618 mm

Parallel to Vessel Wall, opening length d 94.8309 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 24.3364 mm

Weld Strength Reduction Factor [fr1]:

 $= \min(1, S_n / S_v)$ $= \min(1, 135 / 108)$ $= 1.000$

Weld Strength Reduction Factor [fr2]:

 $= \min(1, S_n / S_v)$ $= \min(1, 135 / 108)$ $= 1.000$

Weld Strength Reduction Factor [fr3]:

 $= \min(fr2, fr4)$ $= \min(1, 1)$ $= 1.000$

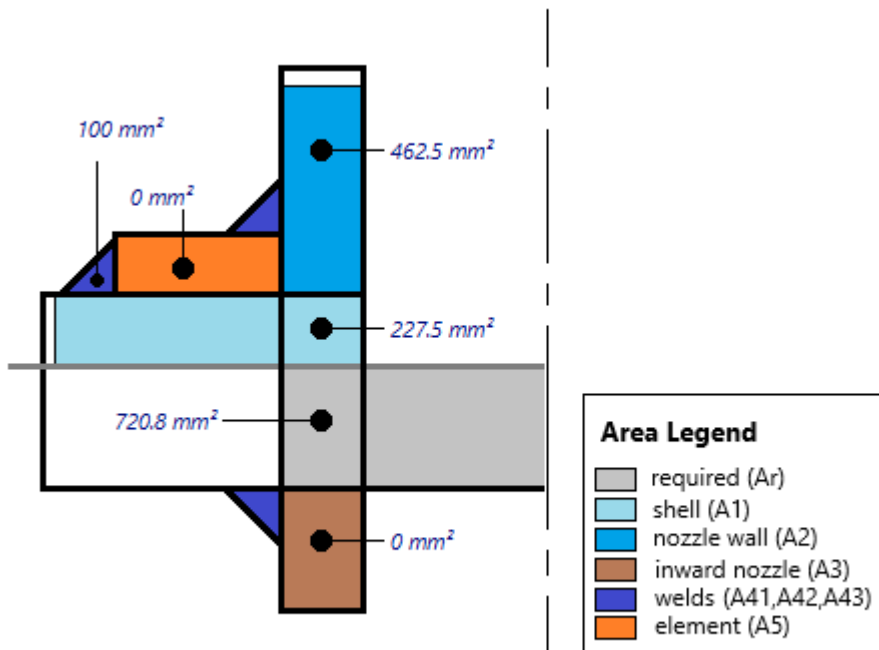


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	720.839	426.212	NA
Area in Shell	A1	227.470	95.885	NA
Area in Nozzle Wall	A2	462.547	424.951	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	790.017	620.837	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (94.8 \cdot 7.6 \cdot 1 + 2 \cdot 9.73 \cdot 7.6 \cdot 1 \cdot (1 - 1)) \\
 &= 720.839 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 94.8(1 \cdot 10 - 1 \cdot 7.6) - 2 \cdot 9.73 \\
 &\quad (1 \cdot 10 - 1 \cdot 7.6) \cdot (1 - 1) \\
 &= 227.470 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 24.3)(9.73 - 0.23)1 \\
 &= 462.547 \text{ mm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10^2 \cdot 1 + (0)^2 \cdot 1
 \end{aligned}$$



= 100.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 1.0038$ mm
 Wall Thickness per UG16(b), $tr_{16b} = 1.5000$ mm
 Wall Thickness, shell/head, internal pressure $trb_1 = 7.6013$ mm
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 7.6013$ mm
 Wall Thickness, shell/head, external pressure $trb_2 = 1.3871$ mm
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 1.5000$ mm
 Wall Thickness per table UG-45 $tb_3 = 5.2578$ mm

Determine Nozzle Thickness candidate [tb]:

= min[tb_3 , max(tb_1 , tb_2)]
 = min[5.26, max(7.6, 1.5)]
 = 5.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(t_a , t_b)
 = max(1, 5.26)
 = 5.2578 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: K2-4"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 9.7345$ mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 * W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, ($A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)$)) S_v
 = max(0, ($721 - 227 + 2 \cdot 9.73 \cdot 1 \cdot (1 \cdot 10 - 7.6)$)) 108)
 = 58430.90 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

= ($A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot fr_2$) $\cdot S_v$
 = ($463 + 0 + 100 - 0 \cdot 1$) $\cdot 108$
 = 60862.81 N

Weld Load [W2]:

= ($A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)$) $\cdot S_v$
 = ($463 + 0 + 100 + (195)$) $\cdot 108$
 = 81926.72 N

Weld Load [W3]:

= ($A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)$) $\cdot S$
 = ($463 + 0 + 100 + 0 + (195)$) $\cdot 108$
 = 81926.72 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= ($\pi / 2$) $\cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 = ($3.14 / 2.0$) $\cdot 114 \cdot 10 \cdot 0.49 \cdot 108$
 = 95182. N

Shear, Nozzle Wall [Snw]:

= ($\pi \cdot (D_{lr} + D_{lo}) / 4$) $\cdot (Thk - Can)$ $\cdot 0.7 \cdot S_n$



$$= (3.14 \cdot 52.3) \cdot (9.73 - 0) \cdot 0.7 \cdot 135$$
$$= 150886. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$
$$= (3.14/2.0) \cdot 114 \cdot (10 - 0) \cdot 0.74 \cdot 135$$
$$= 179112. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (95182 + 150886) = 246068 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$
$$= (95182 + 0 + 179112 + 0) = 274294 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$
$$= (95182 + 179112 + 0) = 274294 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 246067 N	, must exceed W = 58430 N	or W1 = 60862 N
Path 2-2 = 274293 N	, must exceed W = 58430 N	or W2 = 81926 N
Path 3-3 = 274293 N	, must exceed W = 58430 N	or W3 = 81926 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.57 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.0891 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1501.0891 mm

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Input, Nozzle Desc: K3-2''

From: 30

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2449.9998	mm
Shell Finished (Minimum) Thickness	t	10.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		4100.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		0.00	°C
---	--	------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		XS	

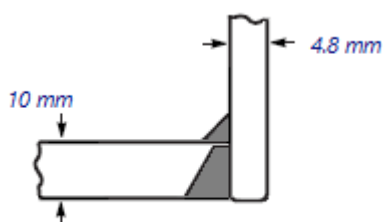
Flange Material		SA-182 F304	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K3-2''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375 in.
Actual Thickness Used in Calculation	0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 30.2 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.1221 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5399 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.2698 mm
Parallel to Vessel Wall, opening length	d	50.6349 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.1126 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2''.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.5399 mm
Wall Thickness per UG16(b),	tr16b	= 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.6013 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.6013 mm
Wall Thickness, shell/head, external pressure	trb2	= 1.3871 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 1.5000 mm
Wall Thickness per table UG-45	tb3	= 3.4200 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.42, \max(7.6, 1.5)]$
 $= 3.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(0.54, 3.42)$
 $= 3.4200 \text{ mm}$

Available Nozzle Neck Thickness = 4.8450 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: K3-2''

Intermediate Calc. for nozzle/shell Welds	Tmin	4.8450 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.3915 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

LYE DRAIN VESSEL

PV Elite 28

Licensee: SL User



FileName : A167NHM07HP001&002&0003_LYE DRAIN VESSELS -7%

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Nozzle Calcs.: K3-2''

Nozl: 38 3:43pm Sep 18,2026



HEXAGON

The Drop for this Nozzle is : 0.3033 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1500.3032 mm

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Input, Nozzle Desc: N4-3''

From: 50

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2600.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		11700.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
---	--	------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		3.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

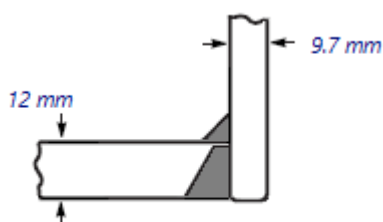
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N4-3"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	3.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$$

$$= 7.6013 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= 0.55 \cdot 44.5 / (135 \cdot 1 + 0.4 \cdot 0.55)$$

$$= 0.1799 \text{ mm}$$

Required Nozzle thickness under External Pressure per UG-28 : 0.7894 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	138.8618 mm
Parallel to Vessel Wall, opening length	d	69.4309 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	24.3364 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N4-3".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.7894 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.6013 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.6013 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3871 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 4.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[4.8, \max(7.6, 1.5)]$$

$$= 4.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(0.79, 4.8)$$

$$= 4.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.1,	Allowable	: 134.8 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.4 MPa	Passed
Occasional	: 0.9,	Allowable	: 179.3 MPa	Passed
Shear	: 20.3,	Allowable	: 94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C



Weld Size Calculations, Description: N4-3"

Intermediate Calc. for nozzle/shell Welds Tmin 9.7345 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.72 MPa

The Drop for this Nozzle is : 0.6588 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1502.6588 mm

Input Echo, WRC297 Item 1, Description: N4-3" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	88.900	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 137.64	MPa
Nozzle Hot S.I. Allowable	SNmh 134.82	MPa

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-4799.79	N
Circumferential Shear	Vc	3599.84	N
Longitudinal Shear	Vl	4799.79	N
Circumferential Moment	Mc	720253.69	N-mm
Longitudinal Moment	Ml	936329.88	N-mm
Torsional Moment	Mt	0.108E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.132) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(1506.0(12.0 - 0.0))}$
 $= 532.526 \text{ mm}$

WRC 297 Curve Access Parameters:



Vessel Mean Diameter (D) = 3012.000 mm
 Nozzle Outside Diameter (d) = 88.900 mm
 Vessel Thickness used (T) = 12.000 mm
 Nozzle Thickness used (t) = 9.735 mm
 T / t = 1.233
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.468$$

Nr/P = 0.187
 Mr/P = 0.268
 M0/P = 0.105
 N0/P = 0.174
 MrD/Mc = 0.276
 NrDL/Mc = 0.156
 M0d/Mc = 0.131
 N0DL/Mc = -0.105
 MrD/M1 = 0.253
 NrDL/M1 = 0.152
 M0D/M1 = 0.117
 N0DL/M1 = -0.024

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-51	51	51	-51
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	45	106	144	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-110	110	110	-110
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	-28	86	215	-112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7
Shear Stress Summary	9	9	5	5

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	75	110	216	113



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	5	5	-5	-5
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	22	68	99	-31

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	53	-53	53	-53
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-93	93	93	-93
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	25	105	229	-62

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	7	7	7	7
Shear Stress Summary	4	4	10	10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	106	230	65

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	9	10	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-117	117	117	-117
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-83	46	195	-146

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	96	49	195	153

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	5	5	-5	-5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	2	2	2
Hoop Stress Summary	13	14	1	2

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	52	-52	52	-52
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-101	101	101	-101
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-62	34	173	-134

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	8	8	8	8
Shear Stress Summary	5	5	12	12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress 76 36 174 139

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	6	6	4	4	-2	-2	14	14
Circ. Q (SUS)	-30	30	72	-72	-40	40	146	-146
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-5	-5	17	17	10	10	0	0
Long. Q (SUS)	-57	57	163	-163	-23	23	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	74.1	74.1	72.2	72.2	66.2	66.2	82.1	82.1
Pm+Pl+Q (Total)	74.2	107.7	214.4	112.4	29.1	106.6	228.6	66.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	108.20	Passed
Pm+Pl (SUS)	82.08	162.30	Passed
Pm+Pl+Q (TOTAL)	228.62	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	2	2	2	2	2	2	2
Circ. Pl (SUS)	6	6	4	4	10	10	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	22	22	-15	-15	17	17
Long. Q (SUS)	-65	65	169	-169	-49	49	153	-153
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm+Pl (SUS)	27.3	27.3	23.2	23.2	26.7	26.7	18.5	18.5
Pm+Pl+Q (Total)	94.1	48.5	192.2	152.5	75.7	36.0	171.7	138.8

Nozzle Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	2.00	134.82	Passed
Pm+Pl (SUS)	27.29	202.23	Passed
Pm+Pl+Q (TOTAL)	192.19	404.47	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-6

From: 50

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2600.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		11100.00	mm
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	134.82	MPa
Allowable Stress At Ambient	Sna	137.64	MPa

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)	Outside		
Layout Angle	90.00	deg	
Diameter	6.0000	in.	

Size and Thickness Basis	Minimum		
Nominal Thickness	80		

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

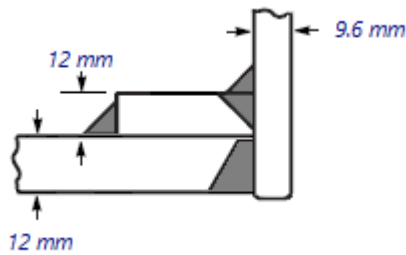
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304L	
Pad Allowable Stress at Temperature	Sp	115.00	MPa
Pad Allowable Stress At Ambient	Spa	115.00	MPa
Diameter of Pad along vessel surface	Dp	300.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		65.8625	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N5-6**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 84.1 / (135 \cdot 1 + 0.4 \cdot 0.55)$
 $= 0.3406 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3893 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	298.1452 mm
Parallel to Vessel Wall, opening length	d	149.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

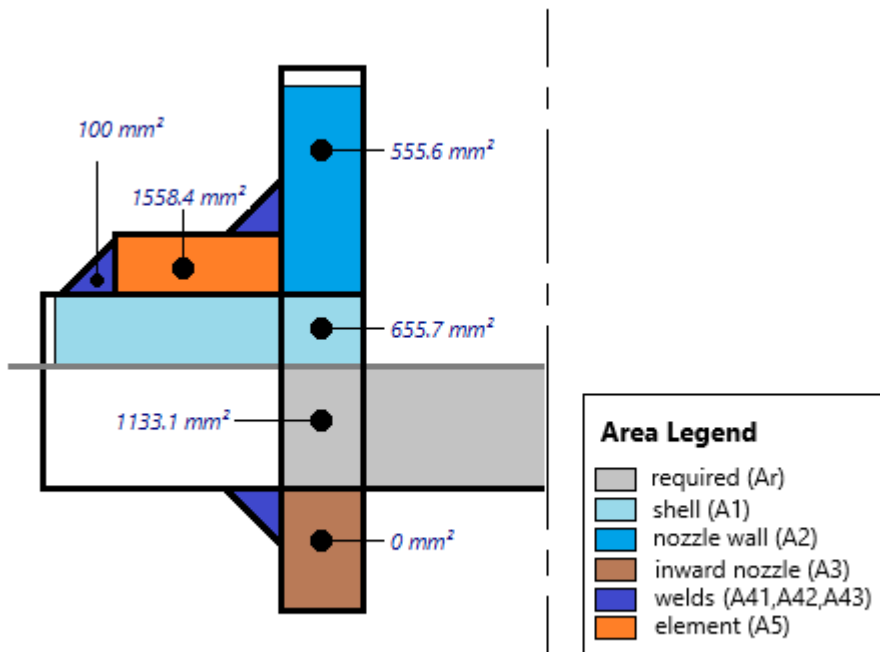


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1133.147	687.836	NA
Area in Shell	A1	655.724	413.199	NA
Area in Nozzle Wall	A2	555.637	492.715	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	1558.442	1558.442	NA
TOTAL AREA AVAILABLE	Atot	2869.803	2564.357	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (149 \cdot 7.6 \cdot 1 + 2 \cdot 9.6 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 1133.147 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 149 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 9.6$$

$$(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 655.724 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 30) \cdot (9.6 - 0.34) \cdot 1$$

$$= 555.637 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot f_{r4}$$

$$= (298 - 168) \cdot 12 \cdot 1$$

$$= 1558.442 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.3893 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.6013 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.6013 mm
Wall Thickness, shell/head, external pressure	trb2 = 1.3871 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 6.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.22, \max(7.6, 1.5)]$$

$$= 6.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.39, 6.22)$$

$$= 6.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.3,	Allowable	: 134.8 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.3 MPa	Passed
Occasional	: 2.0,	Allowable	: 179.3 MPa	Passed
Shear	: 18.2,	Allowable	: 94.4 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N5-6

Intermediate Calc. for nozzle/shell Welds	Tmin	9.6012 mm
Intermediate Calc. for pad/shell Welds	TminPad	12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5 * TminPad	7.0700 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1133 - 656 + 2 \cdot 9.6 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 60791.49 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - C_{an} / 707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (556 + 1558 + 100 - 0 \cdot 1) \cdot 108$$



$$= 239544.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (556 + 0 + 100 + (230)) \cdot 108$$

$$= 95864.74 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (556 + 0 + 100 + 1558 + (230)) \cdot 108$$

$$= 264474.94 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Snw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 148932. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 300 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 249822. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 79.3) \cdot (9.6 - 0) \cdot 0.7 \cdot 135$$

$$= 225826. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 168 \cdot 12 \cdot 0.74 \cdot 135$$

$$= 316431. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 168 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 263692. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (249822 + 225826) = 475648 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (148932 + 316431 + 263692 + 0) = 729055 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (249822 + 263692 + 0) = 513514 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 475648 N	, must exceed W = 60791 N	or W1 = 239544 N
Path 2-2 = 729054 N	, must exceed W = 60791 N	or W2 = 95864 N
Path 3-3 = 513513 N	, must exceed W = 60791 N	or W3 = 264474 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.83 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 2.3616 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1504.3617 mm

**Input Echo, WRC297 Item 1, Description: N5-6 :**

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	168.275	mm
Nozzle Thickness	9.6012	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 137.64	MPa
Nozzle Hot S.I. Allowable	SNmh 134.82	MPa
Diameter of Reinforcing Pad	300.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.55	MPa
Radial Load	P	-9599.58	N
Circumferential Shear	Vc	7199.67	N
Longitudinal Shear	Vl	9599.58	N
Circumferential Moment	Mc	0.288E+07	N-mm
Longitudinal Moment	Ml	0.375E+07	N-mm
Torsional Moment	Mt	0.432E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(1512.0(12.0 - 0.0))}$$

$$= 612.784 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3024.000	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	24.000	mm
Nozzle Thickness used	(t) =	9.601	mm
T / t	=	2.500	
d / t	=	17.526	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.625$$

Nr/P	=	0.141
Mr/P	=	0.156
M0/P	=	0.121
N0/P	=	0.216
MrD/Mc	=	0.206
NrDL/Mc	=	0.161
M0d/Mc	=	0.188



$N\theta DL/Mc = 0.124$
 $MrD/Ml = 0.186$
 $NrDL/Ml = 0.149$
 $M\theta D/Ml = 0.166$
 $N\theta DL/Ml = 0.119$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-38	38	38	-38
Normal Pressure Stress	33	34	33	34
Outplane Stress Summary	6	59	92	-7

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	-13	41	83	-33

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	23	61	93	33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-33	33	33	-33
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-4	38	69	-21

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	33	34	33	34
Inplane Stress Summary	10	53	93	-10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	15	53	94	23

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	3	4	12	13

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-160	160	160	-160
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-124	91	239	-188

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	10	10	10	10
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Shear Stress Summary	12	12	7	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	130	93	239	203

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	4	5	12	12

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-139	139	139	-139
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-98	74	213	-172

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10
Shear Stress Summary	6	6	13	13

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	103	75	214	187

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{1506.0(12.0 - 0.0)}$$

$$= 611.901 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3012.000 mm
Nozzle Outside Diameter	(d) =	300.000 mm
Vessel Thickness used	(T) =	12.000 mm



Nozzle Thickness used (t) = 75.464 mm
 T / t = 0.159
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.578$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.090
 Mr/P = 0.172
 M0/P = 0.050
 N0/P = 0.079
 MrD/Mc = 0.299
 NrDL/Mc = 0.093
 M0d/Mc = 0.092
 N0DL/Mc = 0.062
 MrD/M1 = 0.136
 NrDL/M1 = 0.107
 M0D/M1 = 0.042
 N0DL/M1 = 0.097

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	63	67	123	40

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	68	-68	68	-68
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-70	70	70	-70
Inplane Pressure Stress	34	34	34	34
Inplane Stress Summary	28	32	188	-90

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	63	67	188	131

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-36	36	36	-36
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	34	34	34	34
Outplane Stress Summary	18	51	100	-13

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	68	-68	68	-68
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-119	119	119	-119
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	17	118	268	-107

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	18	118	268	107

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	34	33	34	33	34	33	34
Circ. Pl (SUS)	-1	-1	7	7	-2	-2	6	6
Circ. Q (SUS)	-26	26	50	-50	-21	21	51	-51
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-3	-3	7	7	0	0	6	6
Long. Q (SUS)	-28	28	58	-58	-21	21	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	33.0	34.0	33.0	34.0	33.0	34.0	33.0	34.0



Pm+Pl (SUS)	32.1	33.0	40.1	41.1	31.1	32.1	39.1	40.1
Pm+Pl+Q (Total)	23.3	60.3	90.9	35.3	16.2	53.5	91.0	24.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	108.20	Passed
Pm+Pl (SUS)	41.06	162.30	Passed
Pm+Pl+Q (TOTAL)	91.04	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	68	68	68	68	68	68	68	68
Circ. Pl (SUS)	-3	-3	13	13	-1	-1	11	11
Circ. Q (SUS)	-1	1	41	-41	-51	51	187	-187
Long. Pm (SUS)	34	34	34	34	34	34	34	34
Long. Pl (SUS)	-4	-4	14	14	1	1	9	9
Long. Q (SUS)	-2	2	138	-138	-16	16	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
Pm+Pl (SUS)	65.0	65.0	81.0	81.0	67.0	67.0	79.0	79.0
Pm+Pl+Q (Total)	64.2	66.3	186.0	130.0	19.3	118.0	266.1	108.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	68.00	108.20	Passed
Pm+Pl (SUS)	81.03	162.30	Passed
Pm+Pl+Q (TOTAL)	266.05	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	-1	-1	7	7	0	0	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-19	-19	21	21	-15	-15	17	17



Long. Q (SUS)	-107	107	213	-213	-86	86	192	-192
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	20.4	21.4	23.3	23.4	18.0	19.0	19.9	20.0
Pm+Pl+Q (Total)	129.2	91.6	236.3	202.6	103.9	73.7	211.8	185.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	134.82	Passed
Pm+Pl (SUS)	23.35	202.23	Passed
Pm+Pl+Q (TOTAL)	236.28	404.47	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: M2-24''****From: 50**

Pressure for Reinforcement Calculations	P	0.5466	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2600.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10350.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-240 304	
Material UNS Number		S30400	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	108.20	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

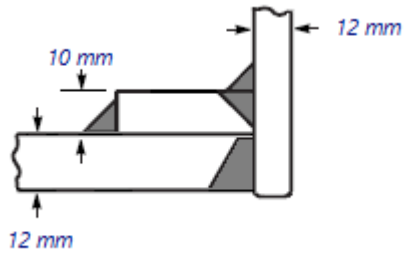
Outside Projection	ho	1490.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304	
Pad Allowable Stress at Temperature	Sp	108.20	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	900.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		145.2000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M2-24''**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.55 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.55)$
 $= 7.6013 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.55 \cdot 305 / (108 \cdot 1 + 0.4 \cdot 0.55)$
 $= 1.5368 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 2.8815 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1171.2000 mm
Parallel to Vessel Wall, opening length	d	585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 108 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

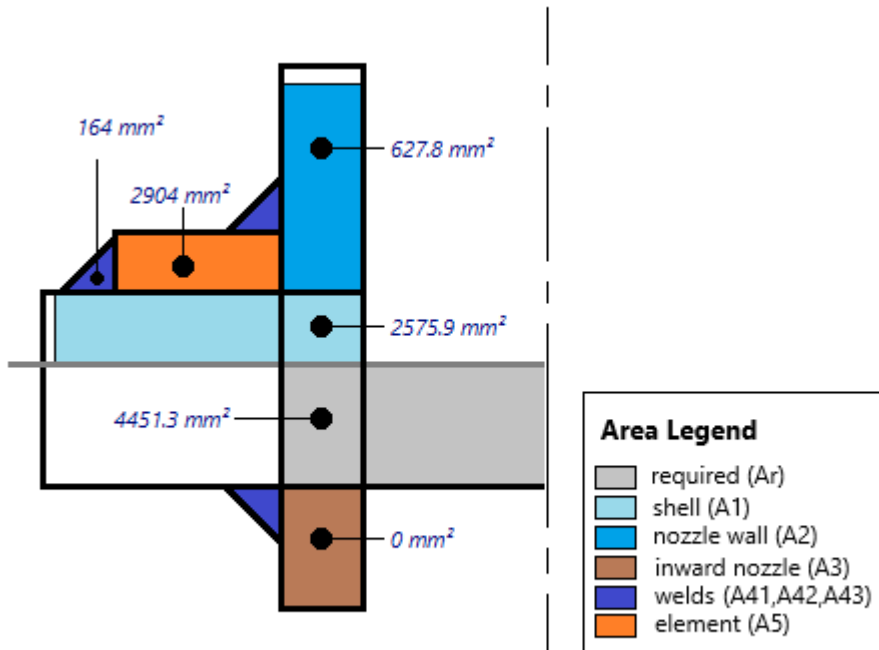


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4451.326	2702.017	NA
Area in Shell	A1	2575.874	1623.166	NA
Area in Nozzle Wall	A2	627.792	547.109	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	164.000	164.000	NA
Area in Element	A5	2904.000	2904.000	NA
TOTAL AREA AVAILABLE	Atot	6271.666	5238.275	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	717.9661	10.0000 mm
Based on given Pad Diameter:	900.0000	3.7316 mm
Based on Shell or Nozzle Thickness:	699.9050	12.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (586 \cdot 7.6 \cdot 1 + 2 \cdot 12 \cdot 7.6 \cdot 1 \cdot (1 - 1))$$

$$= 4451.326 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 586 (1 \cdot 12 - 1 \cdot 7.6) - 2 \cdot 12$$

$$(1 \cdot 12 - 1 \cdot 7.6) \cdot (1 - 1)$$

$$= 2575.874 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot tlwp) \cdot (tn - trn) \cdot fr2$$



$$= (2 \cdot 30) \cdot (12 - 1.54) \cdot 1$$

$$= 627.792 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 8^2 \cdot 1$$

$$= 164.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (900 - 610) \cdot 10 \cdot 1$$

$$= 2904.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 2.8815 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 2.8815 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: M2-24''

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000	mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	5.0000 = $0.5 \cdot T_{minPad}$	5.6560 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (4451 - 2576 + 2 \cdot 12 \cdot 1 \cdot (1 \cdot 12 - 7.6)) \cdot 108)$$

$$= 214329.67 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (628 + 2904 + 164 - 0 \cdot 1) \cdot 108$$

$$= 399853.22 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (628 + 0 + 100 + (288)) \cdot 108$$

$$= 109900.06 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (628 + 0 + 164 + 2904 + (288)) \cdot 108$$

$$= 431012.41 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_o \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 108$$

$$= 507638. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 900 \cdot 8 \cdot 0.49 \cdot 108$$



= 599572. N

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 299) \cdot (12 - 0) \cdot 0.7 \cdot 108$$

$$= 853106. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 610 \cdot 10 \cdot 0.74 \cdot 108$$

$$= 766637. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (10 - 0) \cdot 0.74 \cdot 108$$

$$= 766637. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (599572 + 853106) = 1452678 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (507638 + 766637 + 766637 + 0) = 2040911 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (599572 + 766637 + 0) = 1366209 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1452678 N	, must exceed W = 214329 N	or W1 = 399853 N
Path 2-2 = 2040911 N	, must exceed W = 214329 N	or W2 = 109900 N
Path 3-3 = 1366209 N	, must exceed W = 214329 N	or W3 = 431012 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.66 MPa

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 31.2941 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1533.2942 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

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**Input, Nozzle Desc: N8-4''****From: 50**

Pressure for Reinforcement Calculations	P	0.5768	MPa
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	120	°C

Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	108.20	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3000.00	mm
Design Length of Section	L	2600.0000	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10500.00	mm
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User Entered Minimum Design Metal Temperature		0.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	134.60	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

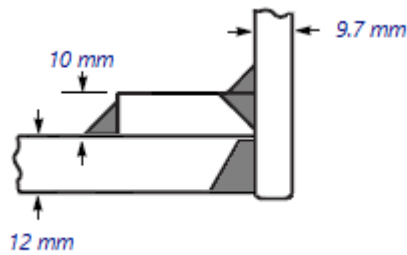
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	240.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-240 304L	
Pad Allowable Stress at Temperature	Sp	115.00	MPa
Pad Allowable Stress At Ambient	Spa	115.00	MPa
Diameter of Pad along vessel surface	Dp	200.0000	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		42.8500	mm

Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N8-4"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.58 \cdot 1500 / (108 \cdot 1 - 0.6 \cdot 0.58)$
 $= 8.0214 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.58 \cdot 57.1 / (135 \cdot 1 + 0.4 \cdot 0.58)$
 $= 0.2445 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5061 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	189.6618 mm
Parallel to Vessel Wall, opening length	d	94.8309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		30.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 135 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 115 / 108)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

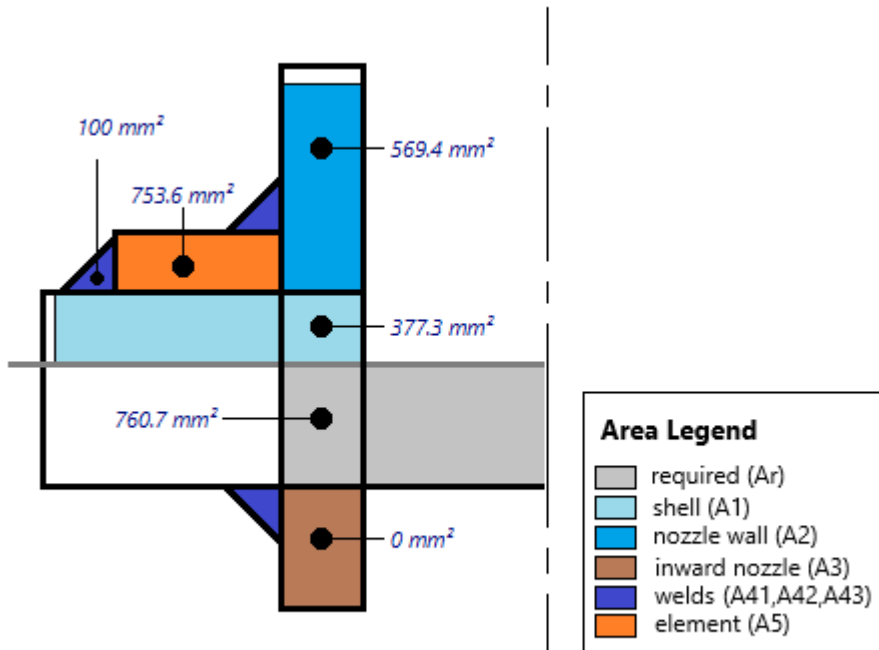


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	760.676	437.559	NA
Area in Shell	A1	377.295	262.852	NA
Area in Nozzle Wall	A2	569.405	553.706	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	753.618	753.618	NA
TOTAL AREA AVAILABLE	Atot	1800.318	1670.176	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (94.8 \cdot 8.02 \cdot 1 + 2 \cdot 9.73 \cdot 8.02 \cdot 1 \cdot (1 - 1))$$

$$= 760.676 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 94.8 (1 \cdot 12 - 1 \cdot 8.02) - 2 \cdot 9.73$$

$$(1 \cdot 12 - 1 \cdot 8.02) \cdot (1 - 1)$$

$$= 377.295 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 30) \cdot (9.73 - 0.24) \cdot 1$$

$$= 569.405 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 0^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (190 - 114) \cdot 10 \cdot 1$$

$$= 753.618 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.5061 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 8.0214 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 8.0214 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 1.3871 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 5.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[5.26, \max(8.02, 1.5)]$$

$$= 5.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(0.51, 5.26)$$

$$= 5.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.5,	Allowable	: 134.6 MPa	Passed
Expansion	: 0.0,	Allowable	: 314.3 MPa	Passed
Occasional	: 1.3,	Allowable	: 179.0 MPa	Passed
Shear	: 19.5,	Allowable	: 94.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

SA-312 TP304, Min Metal Temp without impact per UHA-51: -196 °C

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

Weld Size Calculations, Description: N8-4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.7345	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$5.0000 = 0.5 \cdot T_{minPad}$	$5.6560 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (761 - 377 + 2 \cdot 9.73 \cdot 1 \cdot (1 \cdot 12 - 8.02)) \cdot 108)$$

$$= 49859.01 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (569 + 754 + 100 - 0 \cdot 1) \cdot 108$$



$$= 153958.94 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (569 + 0 + 100 + (234)) \cdot 108$$

$$= 97700.59 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (569 + 0 + 100 + 754 + (234)) \cdot 108$$

$$= 179235.61 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 115$$

$$= 101162. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.14/2.0) \cdot 200 \cdot 8 \cdot 0.49 \cdot 108$$

$$= 133238. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 52.3) \cdot (9.73 - 0) \cdot 0.7 \cdot 135$$

$$= 150635. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.14/2) \cdot 114 \cdot 10 \cdot 0.74 \cdot 135$$

$$= 178814. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 114 \cdot (10 - 0) \cdot 0.74 \cdot 135$$

$$= 178814. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (133238 + 150635) = 283874 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (101162 + 178814 + 178814 + 0) = 458790 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (133238 + 178814 + 0) = 312053 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 283873 N	, must exceed W = 49859 N	or W1 = 153958 N
Path 2-2 = 458790 N	, must exceed W = 49859 N	or W2 = 97700 N
Path 3-3 = 312052 N	, must exceed W = 49859 N	or W3 = 179235 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 0.83 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.0891 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 253.0891 mm

**Input Echo, WRC297 Item 1, Description: N8-4'' :**

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3000.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	MPa
Vessel Hot S.I. Allowable	Smh 108.20	MPa

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	9.7345	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	MPa
Nozzle Hot S.I. Allowable	SNmh 134.60	MPa
Diameter of Reinforcing Pad	200.000	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	0.58	MPa
Radial Load	P	-6399.69	N
Circumferential Shear	Vc	4799.79	N
Longitudinal Shear	Vl	6399.69	N
Circumferential Moment	Mc	0.128E+07	N-mm
Longitudinal Moment	Ml	0.166E+07	N-mm
Torsional Moment	Mt	0.192E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(1511.0(12.0 - 0.0))}$$

$$= 558.662 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3022.000	mm
Nozzle Outside Diameter	(d) =	114.300	mm
Vessel Thickness used	(T) =	22.000	mm
Nozzle Thickness used	(t) =	9.735	mm
T / t	=	2.260	
d / t	=	11.742	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.443$$

Nr/P	=	0.200
Mr/P	=	0.196
M0/P	=	0.136
N0/P	=	0.184
MrD/Mc	=	0.218
NrDL/Mc	=	0.189
M0d/Mc	=	0.186



$N\theta DL/Mc = -0.112$
 $MrD/Ml = 0.205$
 $NrDL/Ml = 0.183$
 $M\theta D/Ml = 0.173$
 $N\theta DL/Ml = -0.049$

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	1	1	-1	-1
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	39	39	39	39
Outplane Stress Summary	22	64	82	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-37	37	37	-37
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	-4	38	80	-24

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	29	65	84	25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	9	39	55	-17

**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-30	30	30	-30
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	39	39	39	39
Inplane Stress Summary	22	52	91	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	23	53	92	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	1	1	-1	-1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	6	7	4	4

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	38	-38	38	-38
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-21	-21	21	21
Axial Bending (ML)	-105	105	105	-105
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-84	48	168	-118

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0



Shear due to Torsion	9	9	9	9
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Shear Stress Summary	12	12	6	6
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	94	51	169	124
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	8	8	2	3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	38	-38	38	-38
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-87	87	87	-87
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-62	36	146	-106
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	9	9	9	9

Shear Stress Summary	5	5	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	71	37	147	113
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{1506.0(12.0 - 0.0)}$$

$$= 557.926 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3012.000 mm

Nozzle Outside Diameter (d) = 200.000 mm

Vessel Thickness used (T) = 12.000 mm



Nozzle Thickness used (t) = 52.585 mm
 T / t = 0.228
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.052$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.104
 Mr/P = 0.221
 M0/P = 0.064
 N0/P = 0.115
 MrD/Mc = 0.311
 NrDL/Mc = 0.072
 M0d/Mc = 0.095
 N0DL/Mc = 0.047
 MrD/M1 = 0.187
 NrDL/M1 = 0.095
 M0D/M1 = 0.059
 N0DL/M1 = 0.110

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	67	74	120	46

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	28	41	169	-77

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	67	74	169	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-25	25	25	-25
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35
Outplane Stress Summary	30	47	85	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-82	82	82	-82
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	72	71	72
Inplane Stress Summary	49	97	221	-61

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear	(Vc)		0	0	0	0
Outplane Shear	(Vl)		-1	-1	1	1
Torsional Shear	(Mt)		2	2	2	2
Shear Stress Summary			0	0	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	49	97	221	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

[illegible]



Pm+Pl (SUS)	42.0	42.0	40.1	40.1	37.1	37.1	45.0	45.0
Pm+Pl+Q (Total)	28.8	64.0	82.9	26.4	22.6	52.6	90.7	18.9

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	39.00	108.20	Passed
Pm+Pl (SUS)	45.04	162.30	Passed
Pm+Pl+Q (TOTAL)	90.68	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-1	-1	11	11	1	1	7	7
Circ. Q (SUS)	-3	3	37	-37	-24	24	140	-140
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	-1	-1	9	9	3	3	7	7
Long. Q (SUS)	-6	6	122	-122	-8	8	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	70.0	71.0	82.0	83.0	72.0	73.0	78.0	79.0
Pm+Pl+Q (Total)	67.2	74.3	166.0	124.0	48.1	97.0	218.1	61.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	108.20	Passed
Pm+Pl (SUS)	83.03	162.30	Passed
Pm+Pl+Q (TOTAL)	218.07	324.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	3	3	1	1	4	4	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-20	-20	22	22	-15	-15	17	17



Long. Q (SUS)	-67	67	143	-143	-49	49	125	-125
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	9	9	9	9	9	9	9	9
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	25.3	25.3	23.2	23.2	21.8	21.8	18.6	18.6
Pm+Pl+Q (Total)	94.6	50.7	166.3	124.8	71.0	36.2	144.0	112.6

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	134.60	Passed
Pm+Pl (SUS)	25.32	201.90	Passed
Pm+Pl+Q (TOTAL)	166.30	403.80	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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